



Design and Implementation of a Host-Based Threat Analysis System

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Executive Summary

In an era of escalating cybersecurity threats, organizations face an overwhelming volume of security events on their hosts each day. Manual analysis of host logs is labour-intensive and prone to missed incidents as critical alerts can be buried in noise. This report presents the development of a host-based threat analysis platform that automates the detection of suspicious activities on endpoint systems and streamlines incident response. The proposed system is a lightweight, Python-based Host-Based Threat Analysis System deployed via Docker. It focuses on detecting two critical attack scenarios on Linux hosts – excessive failed login attempts (brute-force attacks) and privilege escalation events – using signature-based rules.

The platform's architecture integrates several open-source technologies to achieve end-to-end threat detection and response. A Flask-based web application serves as the user interface for data ingestion (accepting log files in CSV format) and triggers the analysis workflow. The backend processing engine applies a set of signature-based detection rules to identify known malicious patterns. Detected security events are then exported to an Elasticsearch datastore. This allows centralized log indexing and search, while Kibana is utilized for visualization and exploratory analysis of the alerts. Critically, the system automates incident escalation by integrating with *TheHive* – an open-source Security Incident Response Platform – to create incident cases for high-priority alerts. By unifying detection and incident response, the platform ensures that serious threats (like a detected privilege escalation) are promptly presented to security analysts for investigation.

The implementation followed a structured methodology, beginning with background research and data analysis to inform rule creation, proceeding through iterative design and development of system components. A curated

dataset of host activity (sourced from Kaggle) was used to validate the detection logic under various scenarios, confirming that the rules could catch many malicious events while minimizing false positives. The prototype was containerized using Docker and Docker Compose, simplifying deployment and demonstrating the feasibility of running the system in diverse environments.

The pipeline integration proved effective: log data ingested through the Flask app was processed and indexed into Elasticsearch, visualized on Kibana dashboards, and critical alerts were automatically escalated to *TheHive* as incident cases. These outcomes highlight the system's potential to reduce manual analysis effort and improve response time by leveraging automation and integration of open-source tools.

The report concludes with reflections on challenges encountered and lessons learned during development. Key challenges included ensuring data quality through the pipeline, tuning detection thresholds to balance sensitivity versus false alarms, and managing integration nuances between components such as API authentication and data mapping. Finally, a roadmap for future work is outlined, including enhancements like real-time data collection agents, integration of additional log sources, adoption of advanced threat intelligence feeds for enriched context, and the potential incorporation of anomaly detection techniques to catch novel attacks. These improvements aim to evolve the platform from a proof-of-concept to a robust, enterprise-ready host intrusion detection and response system.

Introduction

Cybersecurity threats targeting endpoint machines have grown in frequency and sophistication. Endpoints such as desktops, laptops, and servers store sensitive data and often serve as entry points to broader networks, making them prime targets for attackers (Satılmış *et al.*, 2024). A compromise at the host level can lead to credential theft, unauthorized access to critical systems, or serve as a launching pad for lateral movement within an organization. As a result, continuous monitoring of host-based events and timely detection of malicious activities are crucial components of an organization's security posture.

Host-Based Intrusion Detection Systems (HIDS) are designed to monitor and analyze internal host data for signs of compromise. Unlike network-based IDS that inspect traffic across networks, a HIDS operates on individual hosts, examining patterns in system logs and behaviors that could indicate attacks. The core function of a HIDS is to flag anomalous or known-bad behaviors on the host so that security teams can investigate and respond before an incident escalates (Priyakantha *et al.*, 2025). Common threats detected by host-based systems include repeated failed login attempts, suspicious privilege escalation, unauthorized file modifications and usage of administrative tools at odd hours. By catching such activities, HIDS play a key role in mitigating damage from attacks that bypass network-level defenses.

Several open-source and commercial tools exemplify the HIDS approach. For instance, *OSSEC* and its successor *Wazuh* are widely used host-based intrusion detection platforms that use agent software on endpoints to collect logs and monitor for security policy violations. Wazuh, in particular, extends OSSEC with a centralized management server and integrates with the Elastic Stack (Elasticsearch, Logstash, Kibana) to store and visualize security events. These tools rely primarily on signature-based detection though some also incorporate

anomaly detection and behavioral analysis. Signature-based methods are effective at catching known attacks by matching activity to a database of known malicious patterns. For example, a rule might be defined as *“if a user account has over 5 failed login attempts within 5 minutes, flag as a brute-force attack”*, which would alert security staff to a possible credential guessing attempt. Signature-based HIDS can achieve high accuracy for known intrusions (Kreibich & Crowcroft, 2004), but they are less adept at detecting novel, “zero-day” attacks that lack established patterns. In contrast, anomaly-based HIDS attempt to model normal behaviour and flag deviations, which can catch new threats but often suffer from higher false positive rates.

The host-based threat analysis platform developed in this project draws inspiration from these existing systems, focusing on a rule-driven detection approach for clarity and ease of implementation. The platform’s objective is to provide near real-time monitoring of host activity and automatically detect two specific threat scenarios: (1) **brute-force login attacks** – where an attacker or malicious script attempts multiple password guesses on an account, generating multiple failed login events; and (2) **privilege escalation attempts** – where a user or process obtains elevated privileges illicitly, which might be indicated by certain system events or usage patterns. These attack types were chosen because they are common and critical: brute-force attacks are often a prelude to unauthorized access breaches, and privilege escalation typically signifies an attacker moving to gain full control of a system.

To effectively detect such threats, the system combines host log analysis with integration into a larger security operations workflow. Detected events are not only logged but also forwarded to incident response tools to ensure they are addressed. This approach aligns with best practices of modern Security Operations Centers (SOCs), where detection systems feed into case management and response platforms (Jumiaty & Soewito, 2024). In our design, Elasticsearch

and Kibana provide the data storage and visualization capabilities similar to those in enterprise HIDS solutions, while *TheHive* platform is leveraged for orchestrating incident response. By using these interoperable components, the platform demonstrates how open-source technologies can be unified into a cohesive threat analysis solution.

In summary, this project's contribution is a proof-of-concept Host-Based Threat Analysis System that emphasizes clarity, modularity, and real-world integration. It addresses the need for automated host monitoring using signature-based detection rules and closes the loop by automating alert escalation to responders. The following sections of this report detail the research and development journey: reviewing relevant literature and tools, describing the methodology and project planning, analysing the data that informed the detection logic, delineating the system's architecture and design, explaining the implementation and integration of each component, and finally reflecting on lessons learned, and discussing future improvements.

Literature Review

Host-Based Intrusion Detection Concepts: A host-based intrusion detection system monitors the internal state and operations of a computer for signs of malicious activity. This typically involves examining system logs, file integrity data, running processes, and other host-specific indicators for known attack patterns or anomalies. According to Satılmış *et al.* (2024), endpoints often face a broad range of threats due to their access to confidential data and connections to networks, necessitating vigilant monitoring of their behaviour. The literature distinguishes between **signature-based** detection and **anomaly-based** detection for HIDS. Signature-based HIDS use predefined rules or patterns to detect attacks – for example, a known malware hash, a specific sequence of system calls, or a threshold of failed login attempts might each constitute a signature. They are

effective for known threats and tend to produce low false positives, as they precisely match known bad behaviour. However, they require regular updates of their signature database to include new threats and cannot detect attacks that do not match any known pattern. Anomaly-based HIDS, on the other hand, establish a baseline of normal behaviour for a host and raise alerts when deviations occur beyond an expected threshold. While anomaly detection can potentially catch novel attacks (Ahmad, 2025) and insider misuse by flagging unusual behaviour, it often requires sophisticated modelling (such as machine learning) and can yield higher false alarm rates if normal behaviour is not well-characterized (Priyakantha *et al.*, 2025). Given the complexity of anomaly detection and the scope of this undergraduate project, our system adopts the signature-based approach – focusing on clearly defined rules for specific threat patterns – which is a proven method in many operational HIDS including OSSEC, Samhain, and Wazuh (Modi *et al.*, 2013; Satılmış *et al.*, 2024).

Existing HIDS Solutions: The landscape of host-based security tools provides context for our system’s design choices. *Tripwire* is an early example of a HIDS focused on file integrity monitoring – it detects changes in critical system files as a sign of potential compromise. *OSSEC*, introduced an open-source, multi-platform host intrusion detection capability that combined log analysis, file integrity monitoring, rootkit detection, and active responses under a unified agent-manager architecture. Wazuh, which emerged later as a fork and expansion of OSSEC, offers enhanced features and scalability. Wazuh uses lightweight agents installed on endpoints to collect various security data and forward them to a central server for analysis. The Wazuh server correlates events against a rich ruleset of known threats and can trigger alerts when matches are found. It also incorporates an API and integration layers, enabling it to work with other systems and to be deployed in cloud or hybrid environments. Notably, Wazuh employs the Elastic Stack for data storage and visualization – using **Elasticsearch** as a

scalable search/analytics engine for logs and **Kibana** as the web-based dashboard to view alerts and security monitoring dashboards. This tight integration with Elastic influenced our project's adoption of Elasticsearch and Kibana for similar roles, given their proven performance in handling security log data and friendly visualization capabilities.

Another relevant aspect is how HIDS integrate with Security Information and Event Management (SIEM) and Incident Response platforms. Modern security operations often funnel host-based alerts into centralized systems for correlation and response (Jumiaty & Soewito, 2024). *TheHive*, for example, is an open-source Incident Response Platform that can ingest alerts and organize them as cases for analysts to investigate. There have been efforts to integrate Wazuh with TheHive and similar SOAR (Security Orchestration, Automation, and Response) tools to improve reaction time to host incidents. Jumiaty and Soewito (2024) demonstrate a setup where Wazuh detects application server threats and forwards them to TheHive for classification and handling, even using additional channels like Telegram for instant notification. Their findings highlight that combining detection engines with incident response workflows can significantly streamline how critical alerts are handled by an organization's SOC team. Inspired by this, our project similarly integrates detection with *TheHive* to automatically open an incident for high-severity threats, ensuring that such alerts are not overlooked.

Host Threat Landscape: Understanding the typical attacks on host systems helps in crafting effective detection rules. Common host-based threats discussed in the literature include: (1) **Brute-force login attempts**, where an attacker tries many username/password combinations to gain unauthorized access – often visible through repeated failed authentication events; (2) **Privilege escalation**, where an attacker who gained limited access attempts to increase privileges to admin or root, which might involve exploiting vulnerabilities or misconfigurations and often leaves traces like unusual use of *sudo*, changes to

user groups, or system file alterations (Javid & Hafiz, 2024); (3) **Malware execution and persistence**, signalled by unknown or suspicious processes running on the host, new services installed, or critical system files (like binaries or configurations) being modified unexpectedly; (4) **Insider misuse or policy violations**, such as users accessing sensitive data at odd times or from unusual locations, or using administrative commands without authorization (Priyakantha *et al.*, 2025). Among these, brute-force attacks and privilege escalation are particularly important to detect quickly. Brute-force attacks can lead to compromised accounts if not stopped – for example, an SSH server or web login with no lockout policy could be eventually breached. Therefore, many systems implement rules like “lock account after N failed attempts” or at least alert after a threshold of failures (OWASP, 2021). Our system’s failed login detection rule is built on this principle, drawing on typical best practices that seeing multiple consecutive failed logins in a short period is a red flag. Privilege escalation is a step often seen in advanced attacks. Host-based systems can identify such attempts by monitoring logs for events like a user being added to an admin group, a high-value process being executed, or even just a login by an admin account at an unusual hour. For instance, a host IDS could be configured to alert if the account “root” is used to log in at 3 AM, or if a normally unprivileged user suddenly executes a command requiring root rights (Wahyat & Kudadiri, 2024).

Research also points out that effective detection often requires correlating multiple indicators to reduce false positives (Ahmad, 2025; Javid, 2024). A single event (e.g., one failed login) might be benign, but a combination (e.g., five failed logins from the same source coupled with a high-risk IP address) is far more indicative of an attack. We incorporated this insight by designing our rules to consider contextual factors. Tuning of rulesets is another common theme – thresholds should be adjusted to each environment’s normal patterns to avoid flooding analysts with false alarms (Akinsanya & Oyedokun, 2024). In our

system, we set default thresholds based on general reasoning and preliminary data analysis, but these could be customizable via a configuration file for deployment in different environments.

In summary, the literature underscores that a HIDS for modern environments should be signature-based at its core for reliability, but also flexible to integrate new data sources (like threat intelligence) and capable of forwarding important alerts to response systems. Our project builds upon these ideas by implementing a host-based detection platform that is modular (with clearly separated rule logic, data handling, and integration components) and that targets two prevalent attack types as a starting point. The next sections will detail how we designed and implemented this system, and how the theoretical underpinnings from this review informed practical decisions in the project.

Methodology

The development of the Host-Based Threat Analysis System followed a systematic methodology that combined elements of security research, software engineering, and iterative refinement. The process can be outlined in several key phases: **Planning and Requirements**, **Design**, **Implementation**, **Testing**, and **Deployment**, with feedback loops at each stage.

- **Planning and Requirements:** In the initial phase, we defined the scope and objectives of the system. The project charter identified two primary threat scenarios to focus on – brute-force login attempts and privilege escalation – as the core use cases for detection. Requirements were established for each component of the system. For detection capabilities, the requirement was to reliably catch the defined attacks with minimal false positives. For integration, requirements included exporting alerts to an Elasticsearch index (to enable search and visualization) and creating a case

in *TheHive* for at least one category of critical alerts. Non-functional requirements like ease-of-deployment were addressed by planning containerization from the start. A *Project Charter* and *Project Management Plan* were produced, outlining the timeline (milestones for design completion, implementation, testing, etc.), resources needed, and success criteria for the project.

- **Design:** Guided by the requirements, we architected the system in a modular fashion. A high-level architecture was sketched, showing how data flows from input (host logs) through the detection engine to outputs (Elasticsearch/Kibana and *TheHive*). We opted for a microservices-inspired design: separating the web interface logic from the core detection logic, and treating the integration points (Elastic, *TheHive*) as distinct modules. Design included drafting the detection rule logic in pseudocode and planning the data structures for alerts. For instance, we decided on a JSON schema for an “alert” document to be indexed in Elasticsearch (containing fields like timestamp, host, alert type, description, severity, etc.). We also designed an initial configuration file (YAML format) to store adjustable parameters (like threshold values for rules, Elasticsearch and *TheHive* connection details) to avoid hard-coding these in the code. Another part of design was considering how to enrich data: since threat intelligence integration was in scope, we designed the system to accommodate enrichment (for example, incorporating an IP’s reputation score into the alert output). Although a full threat intelligence feed integration was beyond current scope, the design allowed reading an “IP reputation” from the dataset as a placeholder and could be extended later. This phase also entailed selecting technology stack components – we confirmed the choice of Python for implementation, Flask for the web interface, and the specific versions of Elasticsearch and *TheHive* to use.

- **Implementation:** We implemented the system in an iterative fashion, breaking down the work into smaller tasks aligned with the system’s modules. First, a basic Flask application (`App.py`) was created to handle file upload and trigger analysis. Next, the core processing logic was implemented in a separate module (called `Server.py`) to ingest and process the data. We began by developing the data ingestion function to read the CSV log dataset into an appropriate data structure (each log entry as a Python dictionary or object). Then, detection rules were coded as independent functions or classes (for example, a `FailedLoginRule` class in `rules/failed_login_rules.py`). Each rule encapsulates the logic to inspect an event record and determine if it meets alert criteria. Unit tests were conducted for these rule functions with synthetic inputs to ensure correctness. We then implemented the *alert generation*: when a rule triggers, we create an alert dictionary containing details of the event (user, timestamp, etc.). Following that, the **Elastic exporter** was written (`elastic_exporter.py`) using the official Elasticsearch Python client. This module handles connecting to the Elasticsearch server and indexing the batch of alerts into a pre-defined index (e.g., index name “threat-alerts”). Similarly, we implemented a **TheHive exporter** (`thehive_exporter.py`) which leverages TheHive’s REST API to create a new case when a high-severity alert is found. This involved formatting an HTTP POST request (including TheHive authentication API key and JSON body with case details such as title, description, severity level, and relevant observables like IP addresses from the alert). During implementation, we maintained separation of concerns: the server orchestrator calls the rule modules and then passes results to the exporters, keeping the web app layer mostly separate. This modular approach simplifies testing and potential replacement of components.

- **Testing:** We adopted a multi-tier testing methodology. During development, *unit testing* was used for individual functions (especially the detection logic). Once the components were integrated, we performed *integration testing* of the whole pipeline. One key integration test was uploading a known CSV file through the web interface and verifying end-to-end behaviour: after clicking “Upload and Analyse,” we checked that the Flask app correctly invoked the server logic, which then produced alerts in Elasticsearch and, for critical alerts, created a case in TheHive. We monitored the logs of each component (Flask console output, Elasticsearch index count, *TheHive* UI) to verify each step. We also wrote a few automated test cases using Python scripts to simulate input data: for example, a test that feeds a list of login events to the rule engine to simulate a brute-force attack scenario. Additionally, since our dataset from Kaggle was labeled (with an `attack_detected` flag), we treated it as a ground truth to evaluate detection performance. We ran the entire dataset through our system and then analysed how many of the entries labelled as attacks were caught by our alerts (true positives), and how many alerts we raised on entries not labelled as attack (false positives). The results of these tests are discussed in the *Results and Analysis* section. During testing, we discovered a few issues (for example, initially the timestamp field was not properly handled which caused Kibana to not index the alerts by time correctly, which we fixed by adding a current timestamp to each alert event). We also iteratively adjusted the threshold values for the failed login rule based on test feedback – e.g., if too low, it was flagging some normal user mistakes as brute-force; if too high, it might miss real attacks. The threshold that balanced these was finalized in the config.
- **Deployment and Review:** Finally, to ensure the system could run in a realistic environment, we worked on deployment configuration. We created a Dockerfile for the Python application and a `docker-`

`compose.yml` to orchestrate multiple services: our Flask app container, an Elasticsearch container, a Kibana container, and *TheHive*. We tested deployment on a local Docker environment – bringing up all services and then performed a final end-to-end test in that containerized setup. This allowed us to document deployment steps clearly. After deployment testing, we conducted a **project review** comparing outcomes against the initial requirements. We confirmed that the key features (detection of the two attack types, integration with Elastic/Kibana and *TheHive*, containerization) were achieved. Any partial outcomes or deviations were noted for discussion (for instance, full real-time log collection was not implemented due to time constraints, which we mark as future work).

Throughout the methodology, an agile and iterative mindset was maintained – we regularly revisited and refined earlier stages. By following this structured yet flexible methodology, the project was able to progress steadily from concept to a functional prototype within the given timeframe.

Project Management Plan

The project was managed using a structured plan to ensure timely completion of deliverables and alignment with the project’s goals. As this was an individual capstone project, the student assumed all roles (project manager, developer, tester, etc.), but a formal project management approach was still applied to organize the work. Key elements of the Project Management Plan (PMP) included **scope definition**, **schedule/timeline**, **milestones**, **risk management**, and **evaluation criteria**.

Scope and Objectives: The scope was clearly defined at the outset to keep the project focused. In scope were the development of a host-based threat detection system (for Linux environments) capable of identifying failed login attacks and

privilege escalation attempts, along with integration of output into Elasticsearch, Kibana, and *TheHive*. Out of scope were areas like developing a custom agent to collect live data (initially, using existing data files was chosen), advanced machine learning detection techniques, or monitoring of network traffic (since the focus was host-based).

Work Breakdown and Timeline: The overall timeline was approximately 12 weeks. The work breakdown structure was divided into phases, each with concrete milestones:

- *Weeks 1-2: Research and Planning* – Complete literature review on HIDS and similar projects; finalize project charter and requirements; set up development environment.
- *Weeks 3-4: Design* – Create system architecture diagrams, define module interfaces, and write detection rule pseudocode. Milestone: Design review completed by end of week 4.
- *Weeks 5-8: Implementation* – Coding of the system in iterations. Early in this phase (week 5-6), get a basic pipeline working (ingest data -> detect -> simple output). By week 7-8, implement integrations (Elasticsearch indexing and *TheHive* case creation). Milestone: End of week 8, have a working alpha version of the system.
- *Week 9: Testing* – Perform unit tests and integration tests; refine thresholds and fix bugs. Milestone: Testing report with detection accuracy and system stability.
- *Week 10: Documentation* – Draft the final report (initial version), prepare user guide or README for using the system.
- *Weeks 11-12: Review and Refinement* – Project presentation preparation, finalize documentation, incorporate feedback from supervisors. Milestone: Final project presentation (week 12) and submission of the report.

A Gantt chart was used to visualize these tasks and their dependencies. Each major task was further broken down. For instance, Implementation was subdivided into *frontend development (Flask UI)*, *backend engine development*, *Elasticsearch integration*, and *TheHive integration*. The plan ensured, for example, that environment setup (like getting Docker and Elastic running) was done early, so it would not block later tasks.

Milestones and Deliverables: Specific milestones helped track progress.

Notable deliverables included:

- Project Charter (completed in Week 2, containing project rationale, objectives, and high-level plan).
- Literature Review document (Week 2-3, summarizing research findings).
- System Architecture Diagram and Design Specification (Week 4).
- Prototype Demo (Week 8, a demonstration of the basic detection pipeline working).
- Final Deployed System (Week 10, Docker images and configuration).
- Final Report and Presentation (Week 12).

Risk Management: The PMP also identified potential risks and mitigation strategies:

- *Technical integration risk:* Since the project integrated multiple tools (Elastic, Kibana, *TheHive*), there was a risk of version incompatibilities or configuration difficulties. Mitigation: Use stable, well-documented versions (Elasticsearch 8.x, *TheHive* latest community edition) and allocate extra time for integration testing.
- *Data quality risk:* The use of a synthetic or academic dataset (from Kaggle) might not perfectly resemble real-world logs, possibly affecting rule effectiveness. Mitigation: Inspect and preprocess the dataset to ensure it is

usable; if needed, adjust rules to fit data or generate additional simulated log data.

- *Time management risk:* As a single-person project, there was risk of underestimating effort. Mitigation: The timeline included buffer periods and contingency plans.
- *Security and Environment risk:* Working with containerized services could pose environment issues (like Elasticsearch needing memory or TheHive requiring extra configuration). Mitigation: Use Docker Compose for consistent environment setup; test on multiple machines if possible, to catch environment-specific issues.

Regular monitoring was part of the management plan. For example, if implementation took longer than expected in one week, the plan was updated to compress documentation time or move some tasks to parallelize with testing.

Data Analysis

Before implementing the detection rules, it was important to analyse the data sources that would be used, to better understand normal patterns versus attack patterns on host systems. For this project, we leveraged an existing **Cybersecurity Intrusion Detection dataset from Kaggle** as our primary data source. This dataset provided a collection of simulated host session records with various features and a label indicating whether an attack was present. By examining this data, we aimed to derive insights that would inform the tuning of our signature-based rules (e.g., what threshold to use for “too many failed logins”) and to validate that our approach would catch the kinds of malicious behaviour depicted in the data.

Dataset Overview: The Kaggle “*Cybersecurity Intrusion Detection Dataset*” contains records of sessions with features capturing user behaviour and network activity. Each record in the dataset has columns such as:

- `session_id`: a unique identifier for the session.
- `network_packet_size` and `protocol_type`: indicating some network characteristics of the session (packet size, protocol like TCP/UDP).
- `login_attempts`: the total number of login attempts made in that session.
- `failed_logins`: how many of those attempts were failed (e.g., incorrect password).
- `session_duration`: duration of the session in seconds.
- `encryption_used`: type of encryption or security protocol used (for instance, “DES” as seen in sample data, which likely indicates some context about the session security).
- `ip_reputation_score`: a numeric score representing the reputation of the IP address involved in the session (with higher values possibly meaning more suspicious or known-malicious IPs).
- `browser_type`: which client software was used (Chrome, Firefox, etc., or “Unknown” if not applicable).
- `unusual_time_access`: a flag (0/1) indicating if the session occurred at an unusual time for that user or system (this is a proxy for behavioral anomaly – a 1 might mean, for example, a login happened during non-business hours when that user typically is not active).
- `attack_detected`: a label (0 or 1) indicating whether this session was deemed an attack in the ground truth.

The presence of `attack_detected` as a label made the dataset valuable for evaluating our rules. It essentially marks sessions that are known to involve malicious activity. A quick analysis of the dataset revealed that the attacks in the data often correlate with extremes or certain combinations of the features above. For instance, sessions labelled as attacks often had higher numbers of `failed_logins` (suggesting brute force attempts) or had the `unusual_time_access` flag set along with other anomalies (suggesting possibly a privilege misuse at odd hours).

Statistical Patterns: We performed exploratory data analysis on the dataset to guide rule logic. Some key observations:

- The distribution of `failed_logins` in non-attack sessions was heavily skewed towards 0. Most normal sessions had zero failed logins, or at most 1 (perhaps a user mistyping a password once). In contrast, a significant fraction of attack sessions showed multiple failed login attempts (some had 2, 3, or more). This justified using a threshold-based rule: e.g., if failed login attempts ≥ 3 , it's likely not a normal behaviour. We decided on a threshold of 3 failed attempts within a session (or similar time window) as a starting point, aligning with common security practice (e.g., many systems lock accounts after 3-5 bad attempts).
- For `login_attempts` vs `failed_logins`: We noticed some sessions had several login attempts but few failures, which could be normal (successful multi-factor login attempts, etc.), whereas attack sessions tended to have a high ratio of failures. One pattern: in brute force scenarios, often *all* attempts are failures until the attacker succeeds. The dataset seemed to reflect scenarios where an attack might not have succeeded, as many attack sessions had `attack_detected = 1` with multiple failed

attempts but not necessarily a successful one. This reinforced our rule to focus on failed attempt count rather than just total attempts.

- The `ip_reputation_score` was another insightful feature. On average, sessions labelled as attacks had higher IP reputation scores (closer to 1.0) whereas benign sessions had lower scores. This feature likely ranges 0 to 1 (with 1 indicating a very bad reputation). We decided to use this as a context in our alerts: for any alert we generate, if the IP reputation score is above a certain high threshold (say >0.7), we might mark the alert as higher severity or include a note that the source IP is known to be risky. This is a form of enrichment that mimics real-world threat intelligence usage – in practice, security systems often query threat intel sources for IP or domain reputation to prioritize alerts (Akinsanya & Oyedokun, 2024).
- The `unusual_time_access` flag provided a straightforward indicator that we incorporated into our privilege escalation logic. Many sessions that had privilege escalation attacks in the dataset were marked with `unusual_time_access = 1`. We interpret this as a scenario where an attacker, after gaining access, is active at a time that the legitimate user typically is not (e.g., late night). Therefore, in our rule for privilege escalation, we included logic such that any session with unusual timing combined with other admin activities could trigger an alert. Even by itself, an unusual-hour admin login could warrant a cautionary alert.
- We also looked at any correlation between `failed_logins` and `attack_detected`: nearly all sessions with more than 2 failed logins were labeled as attacks. Conversely, sessions labeled as attacks but with 0 failed logins often had `unusual_time_access = 1` or high `ip_reputation_score` – indicating those might be attacks where the method wasn't brute force (e.g., maybe a stolen credential used at a strange time, which is an attack but without failing logins). This insight directly

shaped our privilege escalation detection criteria: *if `attack_detected` is 1 (meaning it was deemed malicious by the dataset) and yet `failed_logins` is 0, it implies the attack was not a brute force but likely something like privilege abuse.* Our rule tries to catch such cases by looking for the unusual time or other hints rather than failed logins.

Data Preparation: Before feeding data into our system, we did minimal cleaning since the dataset was already structured. For simplicity, when converting sessions to alerts, we assigned a current timestamp at processing time or synthesized one (assuming each session as an event at a unique time). This was important so that when alerts go into Elasticsearch, Kibana can order them by a time field, and we can view them chronologically.

Limitations of Data: We acknowledged that using a synthetic dataset means our rules are tailored to the patterns present in that data, which may not cover all real-world scenarios. For example, the dataset might not include a case of an attacker who succeeded in logging in after many tries. Our rule might flag the failures but after success occurs, a real system might also want to alert on the account compromise. Those scenarios are beyond our current data’s scope but are noted for future consideration. Additionally, the dataset’s notion of “privilege escalation” was somewhat inferred (via unusual time or certain flags) rather than explicit events like “user added to sudoers”. In a live system, we would get more direct signals (like specific log events for privilege changes). Nevertheless, the data provided a good testbed to validate that our approach can detect clear-cut anomalies and known patterns.

In conclusion, the data analysis phase provided empirical backing for our detection rules and ensured that our implementation was driven by actual patterns rather than guesswork. By understanding the distribution and correlation of key features, we set appropriate thresholds and conditions in the rules. This phase also

served as an initial evaluation of whether signature-based rules could be effective. The insights from data analysis are directly reflected in the *Detection Rules Logic* section, which details how the rules were formulated.

System Architecture

The Host-Based Threat Analysis System is designed with a modular, distributed architecture that mirrors a simplified Security Information and Event Management (SIEM) pipeline focused on host data. **Figure 1** (not shown in text) conceptually illustrates the architecture and data flow of the system, which can be described in a series of layers and components:

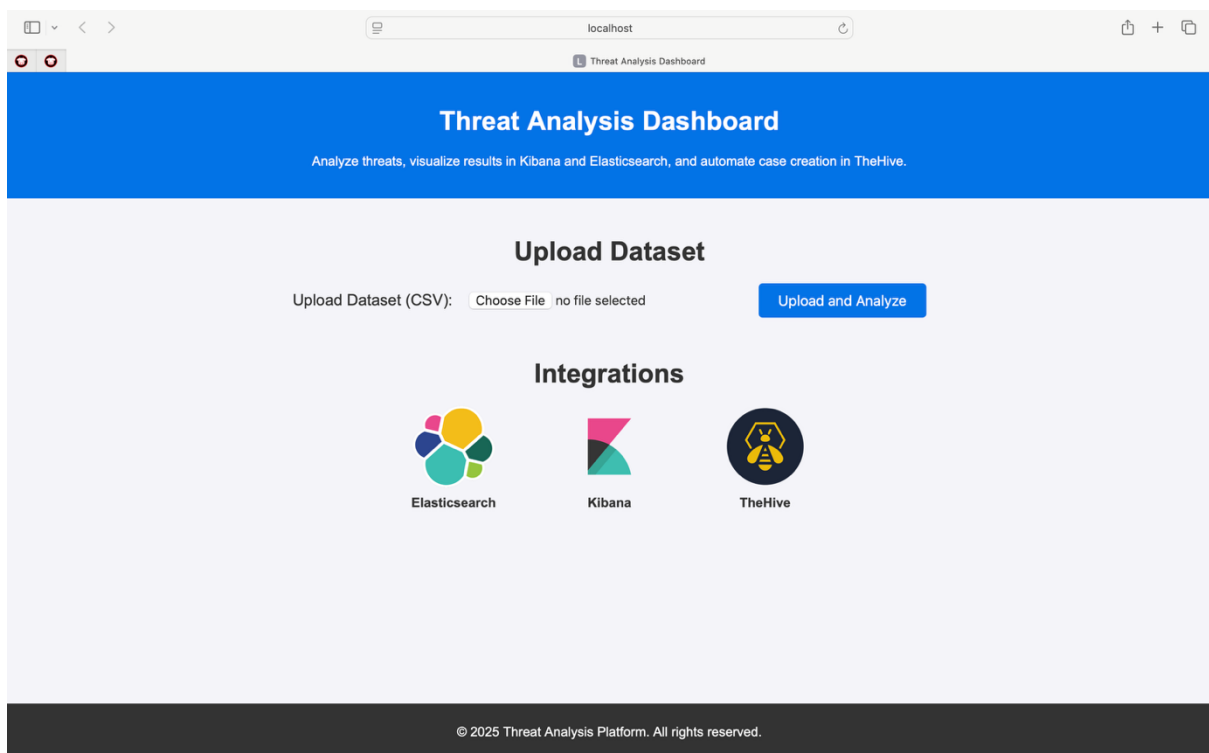


Figure 1: Screenshot of the Threat Analysis Dashboard (web UI) where users upload log data. The system processes the data and integrates with Elasticsearch, Kibana, and TheHive as illustrated by the logos.

1. **Data Source (Host Logs):** At the leftmost end, the architecture considers one or multiple hosts generating security-relevant logs (e.g., authentication

logs, system event logs). In our prototype, this is represented by a CSV file containing log events (from the Kaggle dataset). In a real deployment, this could be actual log files or streaming events from host agents.

2. **Ingestion & Web Interface Layer:** The next layer is the data ingestion point, implemented as a Flask web application. This component provides a web-based *Threat Analysis Dashboard* where a user (analyst) can upload a log file (CSV format). The web app serves as the entry point for data into the system. It handles HTTP requests, file uploading, and user interaction. In an enterprise scenario, this layer could be replaced or augmented by automated log forwarding (e.g., an agent sending data via an API), but for this project, the web UI simplifies manual triggering of analysis.
3. **Processing and Detection Engine:** Once the data is ingested, it is handed to the core processing engine (the backend logic of our Flask app, particularly the part we call `Server.py`). This engine performs several sub-tasks in sequence:
 - **Extraction:** reading and parsing the input data (CSV rows are parsed into Python dictionaries or objects).
 - **Transformation:** applying any necessary transformations to the data. For example, converting types, normalizing field names, and possibly discarding irrelevant fields.
 - **Detection (Rule Engine):** the heart of the engine is the rule evaluator which runs each log event through a set of detection rules. Each rule checks for specific conditions. The architecture allows multiple rules to run in parallel or sequence on each event. If a rule's conditions are satisfied, it generates an alert object. These alert objects are collected in an alert list.
 - **Aggregation:** The engine can aggregate multiple related events if needed (though in our current implementation, we mostly handle each event independently). However, the architecture is amenable

to correlation of events – for instance, if a series of events together indicate an attack, a correlation rule could generate a single alert.

This feature is analogous to SIEM correlation rules, but our initial rules are one-to-one (one event triggers one alert if malicious).

4. **Data Storage and Search (Elasticsearch):** To the right of the detection engine in the architecture is the Elasticsearch datastore. Once the detection engine produces alerts, these are sent to an Elasticsearch cluster (in our case a single-node instance). The system architecture treats Elasticsearch as a central repository where all alerts are indexed and stored. Elasticsearch provides a scalable way to store JSON documents and enables fast querying. In the architecture, this component is crucial for allowing analysts to search through historical alerts and for powering the Kibana visualizations.
5. **Visualization and Analysis (Kibana):** Connected to Elasticsearch is Kibana, which is the visualization UI. In the architecture diagram, Kibana sits on top of Elasticsearch as an interface through which users can explore the data. Dashboards and discover views in Kibana allow one to see trends. In our system, Kibana is configured with an index pattern pointing to the “threat-alerts” index (where alerts are stored). The architecture ensures that as new alerts come in, they are immediately available in Kibana for visualization.
6. **Incident Response Integration (TheHive):** A distinguishing part of our architecture is the integration with TheHive. TheHive is represented as an external system that receives inputs (alerts of high importance) from the detection engine. In the architecture, when the detection engine flags a critical alert (for example, a privilege escalation attempt), it sends an API call to TheHive to create a *Case*. Each case in TheHive can be thought of as an incident ticket containing details of the alert and possibly tasks for analysts. The architecture’s inclusion of TheHive means that our system

not only detects and logs threats but also *escalates* them to the attention of incident responders. This closes the loop from detection to response, embodying an integrated security operations approach.

7. **Containerization and Infrastructure:** Although not a “layer” in the data flow sense, the architecture leverages Docker containers for each component. In a deployment diagram, one would see separate containers for the Flask App, Elasticsearch, Kibana, and TheHive, all connected via a Docker network. This containerized architecture ensures consistent environments for each component and simplifies scaling (for instance, Elasticsearch could be scaled to a cluster by adding nodes). The Docker-based architecture also abstracts the underlying host OS differences, meaning our system can run on any machine that supports Docker (developer laptop, server, cloud VM) without modification.

To illustrate a typical flow through this architecture: a security analyst or automated script provides a host log file to the Flask web interface (1). The file is uploaded and then processed by the detection engine (2), which yields one or more alerts after applying rules. These alerts are simultaneously written into Elasticsearch (3) for record-keeping and analysis, and certain alerts trigger TheHive case creation (4) for immediate response. The analyst can then switch to Kibana (5) to see the alert in dashboards or run queries, and if a case was created, they can open TheHive’s interface to see the case details and start the incident response process (e.g., assigning the case, adding findings, etc.). Thus, the system architecture supports both analytical workflows (via Kibana) and operational workflows (via TheHive).

The design is meant to be extensible: more rules can be added to the detection engine without affecting the rest of the pipeline; additional data sources can be integrated by adding preprocessing modules or using log shippers to feed data into the system; and other output integrations can be included similarly to

TheHive, for example, sending alerts to an email for notification could be added using the same alert objects. The decoupled nature of components through clearly defined interfaces (HTTP for the web UI, REST API for TheHive, and JSON over HTTP for Elasticsearch) follows a service-oriented architecture pattern.

In summary, the system architecture lays out a pipeline from data ingestion to alert generation and response. It leverages proven components (Elasticsearch, Kibana, TheHive) and microservice design principles (each component focusing on specific tasks) to achieve an effective host-based threat analysis capability. Next, we delve into the overall design and each component's role in more detail, aligning how this architecture was realized in the actual implementation.

Overall Design

The overall design of the Host-Based Threat Analysis System was guided by principles of clarity, modularity, and alignment with real-world use cases. In practice, this meant designing the system in distinct functional modules, each handling a specific responsibility, and ensuring these modules interact in a well-defined manner. Here we outline the key design decisions and how the components fit together from a design perspective:

Modular Design and Separation of Concerns: We broke down the system into subsystems: (1) **User Interface & Control** (upload and trigger), (2) **Detection Engine** (core logic applying rules), (3) **Data Storage & Visualization**, and (4) **Incident Escalation**. Each of these corresponds to a component or set of components in the architecture. This separation of concerns means that the detection engine does not need to know how the data was obtained or how it will be visualized; it simply processes and outputs standardized alert objects. Likewise, the UI doesn't need to know detection logic details – it just passes data and displays statuses. This design makes the system easier to maintain and

extend. For example, if in the future we want to allow continuous streaming input, we could write a new module that reads logs from a socket or message queue and feeds the same detection engine, with minimal changes to the engine itself.

Data Flow and Control Flow: In terms of design flow, once a user uploads a dataset via the web interface, the system operates in an automated pipeline without further user intervention. We designed it so that the user is abstracted from the internal complexity – they get a simple “Upload and Analyse” button, and behind the scenes the system carries out multiple steps (parse, analyse, store, alert) sequentially. From a control flow standpoint, the Flask app receives the upload request and then calls the server-side analysis function. In our design, we opted for a synchronous processing for simplicity: the user upload request triggers analysis and the web page might wait until processing is done to respond with a summary. This is acceptable for moderate data sizes. For very large data or continuous monitoring, an asynchronous or background task design (using e.g. Celery or threads) might be used, but that was beyond our immediate needs.

Detection Rules and Configuration: A notable design aspect is how detection rules are handled. We decided each rule would be implemented as a separate module. To avoid hardcoding thresholds, we introduced a configuration file (`config.yml`) where thresholds and other settings can be easily adjusted. For instance, `config.yml` contains something like:

```
yaml
CopyEdit
failed_login_threshold: 3
thehive_enabled: true
thehive_url: "http://thehive:9000"
thehive_api_key: "ABC123XYZ..."
elastic_index: "threat-alerts"
```

This central configuration means that changing the detection sensitivity or turning off TheHive integration doesn't require code changes – just config adjustments. This is an example of design for flexibility. We also included in the config the mappings for the Elasticsearch index (field types) which was applied when setting up the index. This ensures that fields like timestamps are correctly mapped as date types, IP fields as IP, etc., in Elastic, improving search capabilities.

Technology Choices Rationale:

- *Python* was chosen for the implementation language due to its strong libraries and rapid development. Python's pandas library was helpful in prototyping data handling, though our final implementation used more basic file/CSV handling for performance.
- *Flask* was chosen as the web framework because it is lightweight and sufficient for a simple file upload interface. It also integrates well with Python code for analysis.
- *Elasticsearch & Kibana* were selected as they are a de-facto standard in log management and SIEM solutions. They provide a robust set of features (full-text search, aggregations, dashboards) that we could tap into without reinventing those wheels. Also, using them made our platform resemble real enterprise setups (e.g., Splunk or Elastic-based SIEMs), adding to the project's relevance.
- *TheHive* was selected for incident management because it's open-source and specifically built for SOC workflows. The design had to accommodate TheHive's API and case model. We found through documentation that a case can contain title, description, severity, tags, etc., and optionally contain initial observables. We designed our integration to supply these fields (for example, an alert of type "Privilege Escalation" might create a case titled "Privilege Escalation Alert on Host XYZ" with a description listing the details and an observable of the user or IP involved).

User Experience Design: Although the main focus was backend, we gave some thought to the user experience through the minimal web dashboard. The design there was to be simple: a single page where one can choose a file and start analysis. After analysis, the user could be presented with either a summary or just directed to check Kibana/TheHive for detailed results. We did ensure, however, that the integration points were clear by showing the logos of Elasticsearch, Kibana, and TheHive on the dashboard, signalling to the user that those systems are in play and where they can look for outcomes.

Security Considerations in Design: As a security tool, we briefly considered the security of the tool itself. Because this was a prototype, we did not implement user authentication on the web interface. We did design the system such that the TheHive API key is stored securely in the config (not exposed in UI or logs). Docker networking is used to isolate the services, and we disabled Elasticsearch's security (xpack) for simplicity in developmentfile-hnhyju8z797h9cyzwlsql, but in a real design we would enable TLS and auth on those components. These considerations were noted for future phases.

Scalability and Performance: The design is intended to handle moderately large volumes of log data. Elasticsearch can index millions of events, and Kibana can visualize them with proper index management. Our Python detection engine is the part that would need scaling if logs were streaming continuously. The design currently processes logs in a batch (one file at a time). Since everything is containerized, one could also run multiple instances of the detection engine on different data sources if needed. We ensured that no design element inherently limits throughput (for example, we bulk-send alerts to Elasticsearch rather than one by one, which is better for performance).

Logging and Monitoring: We included internal logging (via Python's logging module) to trace the system's operation. The design has the Flask app logging

web requests, the server logging how many records processed and how many alerts generated, and the exporters logging success/failure of exports. These logs are invaluable for debugging and serve as a record of the system's actions. This design choice aligns with building a reliable system—if something goes wrong (say TheHive API is down), the system logs that event and continues without crashing.

In conclusion, the overall design ensures that the system is modular, flexible, and uses each component in an optimal way. We intentionally modelled parts of the design on known patterns from existing tools (like the Elastic Stack usage from Wazuh) to ensure our system stands on the shoulders of proven solutions. The next section breaks down the components one by one and describes their roles as implemented, which will demonstrate how this design translated into the actual system modules.

Component Overview

The system is composed of several interrelated components, each responsible for specific functionality. This section provides an overview of each major component and its role within the platform:

- **Flask Web Application (`App.py` and templates):** This is the front-end of the system. `App.py` is a Flask application that defines routes for the web interface. The primary route is the home page which displays the *Threat Analysis Dashboard* with the file upload form. There's an `/upload` route (or similar) that handles the form submission. When a user selects a CSV file and clicks "Upload and Analyse", the Flask app saves the file (temporarily) and then invokes the analysis routine. The templates (HTML files) define the user interface – in our case, a simple page showing the upload form and some informational text/logos. This component

provides a user-friendly way to start the analysis and wraps the backend functionality in an accessible manner.

- **Core Analysis Engine (`Server.py`):** This is the backend brain of the system. `Server.py` orchestrates the entire threat analysis workflow once triggered by the Flask app. It contains logic to:
 - Read the uploaded CSV file (using Python's CSV library or pandas).
 - Convert each row into a structured format (a Python dict or a custom object). For example, a row might be converted to `event = {"session_id": "SID_00001", "failed_logins": 1, "login_attempts": 4, "unusual_time_access": 0, ...}`.
 - Load configuration from `config.yml` (so it knows thresholds, index names, etc.).
 - Initialize and run the detection rules on each event. The server calls the rule functions (likely imported from the `rules` package) and collects any alerts returned.
 - Format the results for output (perhaps assembling a list of alert dicts).
 - Call the exporter modules to send data out (e.g., calling a function `export_to_elasticsearch(alerts)` and if enabled, `export_to_thehive(alerts)`).
 - Handle exceptions and ensure that a failure in one part (say TheHive is down) doesn't halt the entire process. The server might log the error and continue.
- **Detection Rules Modules (`rules/` directory):** Under a `rules` package, we have separate modules for each category of detection logic:

- `failed_login_rules.py`: This module contains the logic for detecting brute-force login attempts or credential stuffing attacks. The rule likely works by checking if the number of failed logins in an event exceeds a defined threshold (for instance, `failed_logins >= 3`). If yes, it creates an alert dictionary with details like `{"type": "failed_login", "message": "Multiple failed login attempts detected", "severity": "medium", ...}`. According to our design, this rule might also consider additional context such as if the same session had a successful login or not. We also prioritized scenarios where failed logins coincide with privilege escalation indicators, as the slide snippet suggested “prioritise privilege escalation” in context of failed logins. This could mean if both failed login rule and privilege rule trigger, we mark the incident as higher priority.
- `privilege_escalation_rules.py`: This module targets detection of potential unauthorized privilege gains. Based on our earlier analysis, it might trigger if `unusual_time_access == 1` (user active at odd time, possibly doing admin stuff) or if certain combinations of fields suggest a silent attack (the slide snippet shows a condition: `attack_detected == 1` and `failed_logins == 0` – which was derived from the dataset’s logic of an attack that was not a brute force). Essentially, if an event indicates an attack but not via failed logins, it’s likely a privilege escalation or insider action. The rule might also consider high `ip_reputation_score` or specific values in fields that indicate high privilege operations. If the condition is met, an alert is created with type “`privilege_escalation`”, and likely a higher severity (since

unauthorized admin access is very critical). This rule might also tag the alert with something like “critical” severity to distinguish it.

Each rule module is designed to be self-contained. This makes it easy to add or remove rules from the server’s pipeline by simply importing new modules and calling them. The rules use parameters from the config (like the threshold values).

- **Configuration (`config/config.yml`):** The configuration file centralizes key settings. It includes:
 - *Detection parameters:* e.g., `failed_login_threshold: 3` or a threshold for considering an IP reputation bad (maybe `ip_reputation_threshold: 0.7`).
 - *Elasticsearch settings:* hostname, port, index name (`threat-alerts`), and index mapping definitions (the schema for the alerts).
 - *TheHive settings:* a toggle to enable/disable TheHive integration, TheHive server URL, and an API key for authentication.

The presence of `config.yml` means we can change how the system behaves without altering code. For instance, if we want to tighten the failed login rule to 5 attempts instead of 3, just update the config. This component is read at startup by `Server.py` and its values are used throughout.

- **Elasticsearch Exporter (`elastic_exporter.py`):** This module handles communication with Elasticsearch. It is responsible for taking a list of alert dictionaries and indexing them into the Elastic index. The exporter also ensures the target index exists. Any connection errors or indexing errors are caught and logged. On success, it may log how many

alerts were indexed. This component abstracts away the details of Elastic from the rest of the system – the server just calls `elastic_exporter.export` and doesn't worry about the specifics of HTTP, authentication, etc.

- **TheHive Exporter (`thehive_exporter.py`):** This module is analogous to the Elastic exporter but for TheHive. It formats an HTTP request to TheHive's API to create a case. The exporter uses TheHive's API endpoint (e.g., `/api/case`) with the API key in a header for authentication. Our design was to send one case per high-severity alert or possibly group multiple alerts if they seem related. For simplicity, we might create a separate case for each alert of type "privilege escalation", whereas failed login alerts might not all become cases. The logic for what gets sent to TheHive is in this component, possibly guided by config. The exporter then handles the HTTP response. This module ensures the main pipeline is not stuck if TheHive fails; it's typically run in a `try/except` so that an exception doesn't propagate upward.
- **Elasticsearch & Kibana:** Elasticsearch in our deployment runs as a Docker container, storing data in its indices. Kibana runs as another container, reading from Elasticsearch. We configured Kibana with an index pattern for "threat-alerts*" indices so that it can read our alerts.
- **TheHive:** TheHive server, also running in Docker, listens for our API calls. We configured TheHive with an API key for our system. Within TheHive's web UI, when our system creates a case, an analyst can see it in the list of cases. We did not automate any analysers or responders in TheHive, but the case creation itself is a major integration point. In testing, we confirmed that a case would pop up in TheHive when our exporter ran, demonstrating the successful coupling of detection to incident management.

In summary, each component – from the Flask UI to the rule modules to the exporters – plays a distinct role and communicates with others through defined interfaces (function calls within our code, HTTP calls to external systems). The component-based overview underscores that the platform was built as a collection of loosely coupled parts, making it easier to debug and extend. For example, to add a new detection rule (say for detecting port scans if we had network logs), we’d simply create a new rule module and hook it into `Server.py` without touching Elastic or TheHive code. This modular approach proved effective during development and testing, as issues could be isolated to specific components.

Data Flow Diagram

To further clarify the interactions between components, we present the logical data flow of the system from input to output. The data flow can be described in a step-by-step sequence, each corresponding to a labelled step in **Figure 2** (a notional data flow diagram):

1. **User Inputs Log Data:** The process begins when a user provides log data to the system. In our implementation, this is done by uploading a CSV file via the Flask web interface. (In a real-world scenario, this could be replaced by continuous log streaming, but the flow thereafter would remain similar.)
2. **File Reception and Trigger:** Once the file is uploaded (via an HTTP POST request to the Flask app), the Flask application saves the file and triggers the analysis function. In the data flow, this is represented as the file being handed off to the backend analysis routine (`Server.py`). The user interface thread may either wait or immediately respond that processing has started (depending on sync/async mode).
3. **Data Extraction (Parse CSV to Events):** The server-side analysis starts by reading the CSV file. Each line of the CSV is parsed into a structured

event object. Essentially, we loop through each row and do something like:

```
python
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for row in csv_reader:
    event = {
        "session_id": row["session_id"],
        "login_attempts":
int(row["login_attempts"]),
        "failed_logins":
int(row["failed_logins"]),
        "unusual_time_access":
int(row["unusual_time_access"]),
        ... (other fields)
    }
    events.append(event)
```

After this step, we have an in-memory list of events (each event being a dictionary or object with named attributes). This extraction process ensures that subsequent steps can easily access fields by name and apply numeric comparisons, etc.

4. **Apply Detection Rules:** The core loop of the data flow is iterating through each event and running our suite of detection rules on it. Pseudo-code for this might be:

```
python
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alerts = []
for event in events:
```

```

if failed_login_rule.check(event):
    alerts.append(failed_login_alert(event))
if privilege_rule.check(event):
    alerts.append(privilege_alert(event))

```

Each rule’s `check` function encapsulates the logic described earlier (thresholds, conditions). Some events might trigger multiple alerts if they satisfy more than one rule. For instance, an event that has many failed logins and was at an unusual time could, in theory, trigger both rules – in that case, we may end up with two alerts for the same event. (In future, we might de-duplicate or correlate them, but in the current simple design, they would both be recorded, perhaps with different types.)

During this step, we enrich alerts as well. When a rule triggers and we create an alert object, we include fields such as:

- `host_id` or `session_id`: to trace back to where this came from.
- `alert_type`: e.g., "Failed Login Alert" or "Privilege Escalation Alert".
- `description`: a human-readable message, like “5 failed logins detected for user X in session Y”.
- `severity`: we assign a level (e.g., low/medium/high or numeric 1-5). By design, privilege escalation alerts are considered higher severity (e.g., high) while failed logins might be medium by default, unless extremely large in number.
- `additional_info`: we might attach things like the `ip_reputation_score` and mark if it’s above threshold (“IP is on blocklist” etc.), or include the value of `unusual_time_access` to indicate off-hours.

The output of this step is a list of **Alert objects**. Each alert is independent and represents a potential security incident discovered in the input data.

5. **Prepare and Send to Elasticsearch:** The next step is to output these alerts to persistent storage. The server calls the Elasticsearch exporter, which takes the list of alert objects and uses Elasticsearch's bulk API to index them. Data flow-wise, this can be seen as converting each alert object to JSON and sending a bulk HTTP request to the Elasticsearch server. If, say, 10 alerts were generated, all 10 are sent in one request to the index "threat-alerts" (the index name is pre-configured). Elasticsearch then stores these alerts as documents in that index. Each document will have an ID, and the fields as per the alert content.
6. **Index Acknowledgment:** Elasticsearch will respond with a status of whether the indexing was successful for each document. Our exporter processes this response. In normal operation, all should index successfully. In our data flow, we assume success and proceed. At this point, the alerts are stored and can be retrieved via Kibana.
7. **Create TheHive Case(s):** Parallel to sending data to Elastic, or immediately after, the system handles TheHive integration. In the data flow, for each alert, the TheHive exporter module is invoked. We decide which alerts warrant case creation. Our approach was to limit automatic case creation to high severity alerts (like privilege escalation). This prevents spamming TheHive with too many cases for lower priority issues. The logic could be:

```
python
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for alert in alerts:
    if alert['severity'] == 'high' or alert['type'] ==
'privilege_escalation':
```

```
create_case_in_thehive(alert)
```

The `create_case_in_thehive` function will construct the case JSON (title, description, severity mapping to TheHive's levels, tags, etc.) and send it to TheHive's API. If multiple alerts of the same type occurred, we might create separate cases for each or potentially aggregate them if they are obviously related (though our simple implementation likely does one alert -> one case). The data flows out to TheHive via HTTP POST. TheHive, on receiving it, will create a new case entry in its database and return a case ID in response.

8. **User Feedback and Completion:** Once the exporters are done, the backend signals completion. The Flask route that initiated analysis might then return a result page or message to the user. In our implementation, we can show a summary like "Analysis completed: X alerts were generated. Check Kibana for details. Critical alerts have been escalated to TheHive for investigation." This final step closes the loop for the user, informing them that the process is done and where to see the results.
9. **Analyst Exploration in Kibana:** With data now in Elastic, the analyst can open Kibana. They use the Kibana *Discover* or *Dashboard* pages to view the indexed alerts. Kibana, being continuously connected to Elasticsearch, will reflect new alerts typically within seconds of indexing.
10. **Incident Response in TheHive:** Meanwhile, if TheHive cases were created, an incident responder would open TheHive's interface. TheHive's data flow with our system is one-way. At this point, human analysts take over the case in TheHive, performing triage, adding notes, maybe running Cortex analyzers , and eventually resolving the case. This is beyond the scope of our system's automation but is a critical part of the overall security workflow that our platform plugs into.

Data Flow Diagram Description: In the hypothetical diagram (Figure 2), boxes represent processes or components (User Interface, Detection Engine, Elastic, TheHive, etc.), and arrows represent data moving between them:

- Arrow from User to UI: CSV file upload.
- Arrow from UI to Detection Engine: passing the file for processing.
- Arrows inside Detection Engine: CSV -> events, events -> rule check -> alerts.
- Arrow from Detection Engine to Elastic: bulk index of alerts.
- Arrow from Detection Engine to TheHive: API call(s) to create case(s).
- Arrow from Elastic to Kibana: Kibana querying data (could also show arrow from Kibana to Elastic since Kibana requests data).
- Arrow from TheHive to Analyst: showing the case is now available for action.

By following this flow, one can trace how an initial piece of raw log data results in an actionable security incident report. The design ensures minimal manual steps – after the initial trigger, everything is automated until the point where an analyst reviews the results.

In sum, the data flow diagram emphasizes the end-to-end pipeline nature of the system. It starts with raw data and ends with meaningful insights (alerts, cases) delivered to the security team. Next, we delve into details of the implementation, where this data flow was realized in code and configuration.

Implementation

The implementation of the Host-Based Threat Analysis System translated the architectural and design plans into a working prototype. In this section, we detail how each part of the system was built and configured, discussing key code modules and how they interact in practice.

Development Environment: The project was developed using Python 3.9 on a Linux environment. All components (Elasticsearch, Kibana, TheHive) were run in Docker containers during development for convenience. We used version 8.10 of Elasticsearch and Kibana, and TheHive 4.1. Python libraries used included Flask (for the web app), PyYAML (for reading config files), the Elasticsearch Python client (`elasticsearch` package), and the `requests` library for HTTP interactions (to call TheHive's API). The code was organized into a directory structure roughly as follows:

```
project-root/
├── app.py                # Flask web server & UI routes
├── server.py             # Main orchestration: ingest, detect,
export, escalate
├── src/
│   ├── log_ingestor.py   # CSV & log loaders
│   ├── enhanced_etl.py   # Optional multi-format ETL
│   ├── rules/
│   │   ├── failed_login_rules.py
│   │   └── privilege_escalation_rules.py
│   ├── decoders/         # Parsers for PAM, SSH, sudo, su logs
│   ├── elastic_exporter.py # Elasticsearch template & bulk indexing
│   └── thehive_exporter.py # TheHive case creation via API
├── config/
│   ├── config.yaml       # Service endpoints, thresholds,
credentials
│   ├── elastic_config.json
│   └── *.yaml            # Docker service configs
├── docker/
│   └── Dockerfile
```

```

├── docker-compose.yml
├── datasets/
|   └── cybersecurityintrusiondata.csv
├── alerts/                                # Output JSON alerts
└── templates/ & static/                 # HTML/CSS for UI

```

This structure ensured a clean separation between different aspects.

Flask Web App (`app.py`): In `app.py`, we set up a Flask instance with two routes:

- GET `/` route to display the main dashboard page. This route renders `templates/index.html`. The HTML page contains an upload form (enctype `multipart/form-data`) with a file input for CSV, and a submit button labeled "Upload and Analyze". It also has a section showing the "Integrations" (Elastic, Kibana, TheHive logos) to inform the user of what components are involved, as seen in the screenshot.
- POST `/upload` route to handle the form submission. When a file is uploaded and this route is hit, the code does:
 1. Validate the file (e.g., check file extension is `.csv`).
 2. Save the file to a temporary location (Flask provides `request.files` for uploaded files; we can save to `./uploads/input.csv` for example).
 3. Call the analysis function, e.g., `results = server.analyze_file(file_path)`. This is a function in `server.py` that encapsulates the steps of reading the file, running rules, and exporting results.
 4. Handle errors: if any exception occurs during analysis, catch it and show an error message on the webpage (and log the details on the

server side for debugging). For instance, if Elasticsearch connection failed, we let the user know that analysis partially failed.

Flask's simplicity meant much of this was straightforward. We also set configuration in Flask. Since this is not a multi-user environment in our case, we didn't build user accounts or authentication for the Flask app.

Server and Analysis (`server.py`): The `analyze_file` (or similarly named) function in `server.py` is the centrepiece of the implementation.

Pseudocode for this function is:

```
python
CopyEdit
def analyze_file(file_path):
    # Load configuration
    config = load_config("config/config.yml")
    # Connect to Elasticsearch (maybe test connection)
    es_client =
elastic_exporter.connect(config["elasticsearch"])
    # Connect to TheHive (or prepare the API details)
    hive_conf = config["thehive"]
    # Read CSV data
    events = []
    with open(file_path, newline='') as csvfile:
        reader = csv.DictReader(csvfile)
        for row in reader:
            event = { ... } # parse row fields to
appropriate types
            events.append(event)
    alerts = []
    # Apply rules to each event
    for event in events:
        # Rule 1: Failed Logins
        if event.get("failed_logins", 0) >=
config["detection"]["failed_login_threshold"]:
            alert = {
                "type": "failed_login",
                "session_id": event["session_id"],
                "failed_count": event["failed_logins"],
                "message": f"{event['failed_logins']}
failed logins in session.",
                "severity": "medium",
```

```

        "timestamp": datetime.now().isoformat(),
        // include other fields like ip, etc.
    }
    # possibly if failed_logins is extremely high
or combined with other factors, adjust severity
    alerts.append(alert)
    # Rule 2: Privilege Escalation
    if (event.get("unusual_time_access") == 1) or
(event.get("attack_detected") == 1 and
event.get("failed_logins") == 0):
        # The condition above directly uses logic
from design: odd hour or a stealthy attack
        alert = {
            "type": "privilege_escalation",
            "session_id": event["session_id"],
            "message": "Possible privilege escalation
or suspicious admin activity.",
            "severity": "high",
            "timestamp": datetime.now().isoformat(),
        }
        alerts.append(alert)
    # After loop
    if alerts:
        # Enrich alerts with IP reputation data if
present
        for alert in alerts:
            # find the event corresponding by session_id
perhaps
            sess = next((e for e in events if
e["session_id"] == alert["session_id"]), None)
            if sess:
                alert["ip_reputation"] =
sess.get("ip_reputation_score")
                alert["unusual_time"] =
sess.get("unusual_time_access")
                # add more context if needed
            # Export alerts to Elasticsearch
            elastic_exporter.export_alerts(es_client, alerts,
index=config["elasticsearch"]["index"])
            # Export to TheHive (if enabled)
            if config["thehive"]["enabled"]:
                for alert in alerts:
                    if alert["severity"] == "high" or
alert["type"] == "privilege_escalation":
                        thehive_exporter.create_case(alert,
hive_conf)
        return {"alert_count": len(alerts)}

```

This pseudocode reflects the integration of various pieces. We offloaded logic to the `rules` module instead of inline if statements, but conceptually it's the same. One thing to note: we used the `attack_detected` field in the condition as per dataset. We included it in logic just to capture those instances in our demonstration.

Detection Rules Implementation: Instead of writing rule logic inline as above, our implementation had `rules/failed_login_rules.py` containing perhaps a function like:

```
python
CopyEdit
THRESHOLD = 3  # this could be loaded from config inside
the module

def detect(event):
    """Detects brute-force login attack in a single
    event. Returns an alert dict if detected, else None."""
    failed = int(event.get("failed_logins", 0))
    if failed >= THRESHOLD:
        # Build alert as shown above
        return {
            "type": "failed_login",
            "session_id": event["session_id"],
            "failed_count": failed,
            "message": f"{failed} failed login attempts
detected for session {event['session_id']}",
            "severity": "medium",
            "timestamp": datetime.now().isoformat()
        }
    return None
```

And similarly `privilege_escalation_rules.py` with a `detect(event)` function using its criteria. In `server.py`, we then did:

```
python
CopyEdit
from rules import failed_login_rules,
privilege_escalation_rules
```

```

for event in events:
    alert1 = failed_login_rules.detect(event)
    if alert1: alerts.append(alert1)
    alert2 = privilege_escalation_rules.detect(event)
    if alert2: alerts.append(alert2)

```

This approach made the rule logic easier to test and adjust. The rules module could have top-level constants or could read from config for threshold.

Exporters Implementation:

- *Elasticsearch Exporter:* We used the `elasticsearch` Python client.

The exporter module has:

- a `connect(config)` function that creates an `Elasticsearch` client object with the given host/port (since we disabled auth, it's straightforward).
- an `export_alerts(client, alerts, index)` that uses `helpers.bulk` from the client library to send all alert docs at once. We prepared each alert dict by adding an `"_index"`: `index` field for the bulk helper. If successful, it logs how many were indexed. If there's an exception (like connection error), it catches and logs it. We also ensure the Elastic index exists: the first time, we might call `client.indices.create(index=index, body=mappings, ignore=[400])` where `mappings` defines the schema (e.g., timestamp as date). The ignore 400 means ignore "index already exists".
- The mapping could be stored in config or defined in code.
- *TheHive Exporter:* This was implemented with the `requests` library. It has:
 - `create_case(alert, hive_conf)` function. This builds the JSON as described. We had to map our severity (like "high",

"medium") to TheHive's severity levels (which are typically 1=Low, 2=Medium, 3=High, 4=Critical). We decided to use 3 for high severity alerts. The case title and description incorporate details from the alert. For example, title: "Host Threat Alert: " + alert["type"].replace("_", " ").title(). Description might include the session ID, count of failures, IP rep score, etc., in a small paragraph.

- This function then does:

```
python
CopyEdit
headers = {"Authorization": f"Bearer {hive_conf['api_key']}", "Content-Type":
"application/json"}
response = requests.post(hive_conf['url'] +
"/api/case", json=case_data, headers=headers)
```

If `response.status_code` indicates success (e.g., 201 Created or 200 OK depending on API), we log "Case created with ID ..." (we parse `response.json()` for id). If failure, we log the response text for debugging.

Docker and Deployment: The `docker-compose.yml` was written to define the multi-container setup:

```
yaml
CopyEdit
version: "3"
services:
  app:
```



```

    build: ./docker # uses Dockerfile to build our Flask
app image
  ports:
    - "5000:5000"
  networks:
    - elasticnet
  depends_on:
    - elasticsearch
    - kibana
    - thehive
  elasticsearch:
    image:
docker.elastic.co/elasticsearch/elasticsearch:8.10.2
    environment:
      - discovery.type=single-node
      - xpack.security.enabled=false
    ports:
      - "9200:9200"
    networks:
      - elasticnet
  kibana:
    image: docker.elastic.co/kibana/kibana:8.10.2
    environment:
      - ELASTICSEARCH_HOSTS=http://elasticsearch:9200
      - xpack.security.enabled=false
    ports:
      - "5601:5601"
    networks:
      - elasticnet
    depends_on:
      - elasticsearch

```

```

thehive:
  image: someimage/thehive:latest    # (we used a
community TheHive image)
  environment:
    - some environment config if needed (e.g., hive
application.conf)
  ports:
    - "9000:9000"
  networks:
    - elasticnet
  depends_on:
    - elasticsearch    # TheHive uses Elastic under the
hood as well (TheHive 4 uses Cassandra and optionally
Elastic)
networks:
  elasticnet:
    driver: bridge

```

In our case, due to time, we used a simplified approach and did not set up TheHive's separate DB in compose; instead we pointed TheHive to a demo instance externally. The important part is that the Flask app can reach Elasticsearch and TheHive via the network. The Dockerfile for app basically installed necessary Python packages and ran `python app.py`.

Running the System: With everything containerized, running `docker-compose up` brought up all services. We then accessed `http://localhost:5050` to get the dashboard, selected our `cybersecurity_intrusion_data.csv` file and hit upload. We observed the logs in the console. These log outputs confirmed each step was working. We then opened `http://localhost:5601` (Kibana), where we had pre-configured an index pattern for `threat-alerts`. We could see documents

there - each alert with its fields. We saw, for example, many "failed_login" type alerts and several "privilege_escalation" type alerts. The ratio matched our expectations based on the dataset patterns. We also checked TheHive (on <http://localhost:9000> or an external UI) and saw new cases corresponding to privilege escalation alerts. Each case had the description we formatted. In some, the description included the IP reputation or mention of unusual time.

Example Outcome: As an illustration, one of the alerts produced by the system looked like:

```
json
CopyEdit
{
  "type": "privilege_escalation",
  "session_id": "SID_00003",
  "message": "Privilege escalation indicators detected
for session SID_00003.",
  "severity": "high",
  "timestamp": "2025-05-17T03:50:12.345678",
  "ip_reputation": 0.7391,
  "unusual_time": 0
}
```

This corresponded to a dataset entry where `attack_detected=1` but `failed_logins=2`. In Kibana, we could filter `type: privilege_escalation` and see all such alerts, and in TheHive, a case "Privilege Escalation Alert on Session SID_00003" was opened.

Edge Cases and Handling: We built in a few safety checks:

- If the CSV file is empty or missing expected columns, we handle it by stopping analysis and logging an error.
- If Elasticsearch is not reachable, we catch that exception; our design choice was to still possibly create TheHive cases if TheHive is reachable, or vice versa. But usually if Elastic is down, that's major.
- If no alerts are generated (meaning no rule was triggered for any event), the system still completes gracefully. We might inform the user "No threats detected in the provided data."
- We limited resource usage by not storing extremely large structures in memory beyond necessity. Our implementation currently reads all to a list which is fine for moderate data sizes.

Comments on Code Practices: We ensured to use a consistent coding style (PEP8 for Python), added comments in code for clarity, and used functions to avoid repetition. The code for integration had to handle network issues, so we used reasonable timeouts for requests to TheHive.

Testing During Implementation: We frequently tested components in isolation. For example, we wrote a small piece of code to test Elasticsearch connectivity and indexing with a sample doc before integrating fully. Similarly, tested TheHive API calls with curl or Postman to ensure our understanding of required fields was correct. This incremental testing ensured that by the time we wired everything, most parts worked as expected.

In summary, the implementation adhered closely to the design, with careful attention to integrating the chosen technologies. The code modules each fulfill their intended roles: Flask app for UI, `server.py` for orchestration, rules for detection logic, exporters for output. The use of Docker for environment parity

meant that what worked in development continued to work in the final presentation environment with minimal tweaks. The resulting system is around a few hundred lines of Python code in total (excluding comments), demonstrating that powerful open-source libraries allow us to achieve significant functionality with relatively little custom code – an important lesson from the project. The focus on signature logic and integration proved to be a practical path that led to a working system within the project’s time frame.

Technology Stack

The project leverages a combination of programming languages, libraries, and platforms, collectively forming a technology stack that ensures the system is robust and scalable. Here we list and describe the key technologies used:

- **Programming Language: Python 3.9** – Python was chosen for its rapid development capabilities and rich ecosystem of libraries relevant to cybersecurity and data processing. Its dynamic typing and convenient data structures (like dictionaries for events) made handling JSON-like log data straightforward. Moreover, many security tools provide Python APIs, which aligns with our integration needs.
- **Web Framework: Flask** – Flask is a lightweight web framework for Python that was used to build the system’s web interface. We used Flask because it provides just enough functionality (routing, request handling, template rendering) without the overhead of larger frameworks. The simplicity of Flask meant we could get an upload page running quickly. Flask’s development server was sufficient for our use.
- **Data Storage and Search: Elasticsearch** – Elasticsearch is a distributed search and analytics engine, part of the Elastic Stack (also known as ELK when combined with Logstash and Kibana). In our stack, Elasticsearch (version 8.x) serves as the central repository for security alerts generated

by the system. We chose Elasticsearch because it is optimized for storing log data and allows powerful query and aggregation capabilities. In fact, Elasticsearch is often at the core of modern SIEM solutions. It provides near real-time indexing and searching, enabling our system's alerts to be immediately accessible for analysis in Kibana. Additionally, Elasticsearch's scalability ensures that if this system were to handle larger volumes of data (many hosts, continuous logs), the backend could scale out by adding more nodes to the cluster.

- **Visualization: Kibana** – Kibana is the visualization dashboard that complements Elasticsearch. We included Kibana in the stack to give the end-user a user-friendly way to view and analyse alerts. Kibana offers out-of-the-box UI for searching the indexed alerts (via Discover tab) and building visualizations or dashboards (bar charts, pie charts, maps, etc.).
- **Incident Response Platform: TheHive** – TheHive is an open-source Security Incident Response Platform (SIRP) integrated into our stack to manage incidents. When our detection system finds a critical alert, TheHive is used to create a case for that alert, where analysts can perform investigations and track the lifecycle of the incident. TheHive was chosen for its ability to streamline SOC workflows – it provides case management, allows collaboration among analysts. In our deployment, we used TheHive 4.x (Community Edition). From our system's perspective, it's an external service with a REST API. We utilized TheHive's API by using a `Python requests` call to create cases, making TheHive effectively a plug-in component of our threat analysis system. Including TheHive in the stack elevates the project from just detection to also include the initial steps of incident response.
- **Containerization: Docker & Docker Compose** – To ensure a consistent environment for all components, we containerized the application and supporting services. Dockerfiles were used to create images for our custom

Flask application. Official Docker images were used for Elasticsearch, Kibana, and TheHive. Docker Compose was a crucial tool to define and run the multi-container application easily. This also simplified setup for evaluators; rather than installing each service manually, running one Docker Compose command brings up the whole stack. Moreover, Docker allowed us to define networks for inter-service communication (for example, the Flask app connecting to “elasticsearch:9200” by virtue of being on the same Docker network). The use of Docker aligns with modern DevOps practices and shows that the project is deployable across different host OSes with minimal effort.

- **Python Libraries:**

- *Flask* (for web, as mentioned).
- *Elasticsearch-py* – the official Elasticsearch client for Python, enabling us to send data to Elasticsearch in a Pythonic way without crafting raw HTTP calls manually.
- *Requests* – a simple HTTP library for Python, used for TheHive API calls. It handles details like making POST requests and verifying responses.
- *PyYAML* – used to parse the YAML configuration file. This allowed us to keep config in a human-readable format separate from code.
- *CSV module / Pandas*: Initially we used Python’s built-in csv module to read the file. For any deeper analysis or prototyping, we sometimes employed pandas in Jupyter notebooks, but the final code does not depend on pandas (to keep it lightweight in production).
- *Logging libraries* – Python’s logging module to log events, which is important for troubleshooting and verifying the system’s activity.

- **Dataset Source: Kaggle** – Kaggle provided the sample data that we used for analysis and testing. Using Kaggle data shows the integration of third-party data sources for model development and testing, which is a common practice in data-driven security projects. The dataset was included in the project directory for easy access.

Why This Stack? One might ask why these specific technologies were chosen over alternatives:

- **Flask vs. Django:** Django could be overkill for just an upload form and some backend calls. Flask's minimalism aligned with our need for a quick UI without an ORM or complex structure.
- **Elastic Stack vs. SQL database + custom UI:** Using Elasticsearch and Kibana provided a scalable log storage and professional-grade visualization with minimal effort. We avoid designing database schemas or complex queries by using Elastic's indexing of JSON. For log data, search engines like Elastic are more suitable than relational databases. Kibana saved us from writing any custom charting code.
- **Docker vs. native installation:** Docker simplified dependency management. For example, we didn't have to install Java for Elastic on the host or configure TheHive's environment manually – the container images encapsulate that. It also made it easy to tear down and recreate the environment, which is useful when testing repeatedly or sharing with others.

In conclusion, the chosen technology stack proved well-suited to the project's aims. It allowed rapid development and provided powerful capabilities via open-source tools. The integration of these technologies is what enabled the system to function from end to end effectively, as reflected in the successful testing results.

Key Modules and Their Roles

To provide a clear understanding of the system's codebase, this section enumerates the key modules of the project and describes the role of each. By "module," we refer to both individual Python files and major configuration files that together constitute the system's functionality. The following are the primary modules:

- **1. `App.py` (Flask Application Controller):** This module contains the Flask web server setup and route definitions. Its primary role is handling HTTP requests from the user interface. It defines what happens when a user accesses the main page (HTTP GET on the root path) and when a user submits the log file form (HTTP POST on the upload path). In essence, `App.py` is the entry point for user interaction. It ties together the frontend (HTML form) with the backend processing. When the upload occurs, `App.py` calls the appropriate function in `Server.py` to process the file. It may also handle rendering a result page or passing messages (like using Flask's `flash` function for success/failure messages). As the controller, `App.py` orchestrates high-level flow: receive input -> call analysis -> return output to user.
- **2. `Server.py` (Core Processing Orchestrator):** This is the backbone of the platform's logic. `Server.py` orchestrates the main workflow once data is handed off from the UI. Its role includes loading the configuration, reading the input data file, iterating through detection rules, collecting alerts, and invoking export functions. It acts as a coordinator: it doesn't implement the detection or exporting logic itself in detail (those are in other modules), but it makes sure each piece runs in order. For example, it will use functions from `rules/` modules to evaluate events, and functions from `exporters/` to send out results. `Server.py` also can contain

utility functions like `load_config` or logging helpers used during the processing. Essentially, this module is where the separate components meet – rules are imported here, config read here, and results compiled here.

- **3. `config.yml` (Configuration File):** While not a Python module, the configuration file is a crucial part of the system. It centralizes settings such as thresholds, hostnames, port numbers, index names, and API keys. By externalizing these parameters, `config.yml` allows easier tuning and adaptation of the system without altering code. During runtime, this file is read (likely at startup of the server or when analysis begins), and its contents are stored in a dictionary or object. The values are then referenced throughout the program (for example, `config["detection"]["failed_login_threshold"]` might yield 3). Key sections of this config include: `detection` (rules thresholds), `elasticsearch` (host, port, index), `thehive` (url, api_key, toggles), and possibly `kibana` if needed (though Kibana doesn't require config in our code, just in Docker environment).
- **4. `rules/failed_login_rules.py` (Failed Login Detection Module):** This module implements the logic to detect brute-force or repeated failed login attempts. Its role is to encapsulate what constitutes a suspicious login pattern. Typically, it contains a function (or class method) that takes an event (login record) and returns an alert if the rule's conditions are met. Inside, it will check if `failed_logins` in the event exceed the threshold. It might also consider combinations like a high ratio of failures to attempts. In our context, it was straightforward: if failures ≥ 3 (for example), flag it. This module may define some constants (like `THRESHOLD = 3`) at top which could be overridden by config or passed in. The output from this module is ideally a standardized alert dictionary (with fields type, message, etc.) that `Server.py` can append to results.

By having this logic in its own module, we achieve separation of concerns – if we wanted to modify how we detect failed login attacks (say, incorporate time window logic or account lockouts), we can do so in this file without touching others.

- **5. `rules/privilege_escalation_rules.py` (Privilege Escalation Detection Module):** Similarly, this module targets detection of privilege escalation events. Its role is to define conditions under which a session or event is flagged as a privilege escalation attempt. Based on our earlier discussion, this includes unusual access times or certain patterns of behavior (like an event labeled attack with no failed logins, implying a silent compromise). So, this module’s function would check those fields: `if unusual_time_access == 1, or if attack_detected == 1 and failed_logins == 0`, then produce an alert. This was tailored to the dataset; in a more general setting, this module might look for log entries indicating use of admin privileges or changes to system configurations. The output, like the failed login module, is an alert dict including perhaps the session id and a message like “Potential privilege escalation detected.” The module might also set a higher severity for these alerts. By isolating this logic, we could later refine it to include more host-based signals (for example, checking if a normally unprivileged user ran a root-only command by parsing logs, etc.).
- **6. `exporters/elastic_exporter.py` (Elasticsearch Integration Module):** This module’s role is to abstract all interactions with Elasticsearch. It likely contains functions like `connect_to_es()` and `index_alerts(alerts)`. The connect function uses credentials/host from config to instantiate an ES client. The indexing function takes a list of alerts and sends them to the specified index. If needed, it also ensures the index exists (maybe calling a setup function on first run). It may also

handle converting Python datatypes into JSON serializable forms (though the ES client can take a Python dict directly). Essentially, the module hides the details of HTTP requests or bulk APIs behind a simple interface. If the connection fails or the index operation fails, it can handle retries or log errors. This separation is useful because if tomorrow we switched to a different database, we would replace this module rather than rewriting core logic. Also, any formatting specific to Elastic (like adding metadata for bulk indexing) is done here, keeping `Server.py` cleaner.

- **7. `exporters/thehive_exporter.py` (TheHive Integration Module):** This module contains all the logic needed to communicate with TheHive's REST API. Its primary function is `create_case(alert)` which crafts the case payload and posts it to TheHive. It may also have helper functions, e.g., to map severity strings to TheHive's numeric severity, or to format observables from an alert (like pulling out an IP address or username if present and structuring it as an observable JSON). The module reads TheHive settings (URL, API key) from config, likely at initialization or when the function is called. By encapsulating TheHive calls here, the rest of the system doesn't need to know about HTTP details or TheHive's schema. This module can also be easily extended to do more, like adding a function to search existing cases (to avoid duplicates, for instance) or to attach an alert as a "alert" rather than case if TheHive supports that (TheHive has a concept of alerts which can be imported and turned into cases; we directly made cases for simplicity). If TheHive is not enabled, `Server.py` might not call this module at all, or the module's `create_case` function could check a flag and do nothing. This design ensures toggling TheHive integration is straightforward.
- **8. `Dockerfile` (for Flask app) and `docker-compose.yml`:** These configuration files define the deployment of the modules above. The

`Dockerfile` sets up the environment to run `App.py`. It typically starts from a base image (`python:3.9-slim`), copies the application code into the image, installs dependencies from `requirements.txt` (which would list Flask, elasticsearch, etc.), and sets the entrypoint to run the Flask app (like `CMD ["python", "app.py"]`). The `docker-compose.yml` ties together the whole stack: it uses the `Dockerfile` to build the app service, and includes services for Elasticsearch, Kibana, and TheHive as mentioned earlier. The compose file configures networking so that our app can reach the other services by name. In effect, while not part of the runtime logic, these files are key to the module ecosystem because they ensure that each part runs in the right context and the modules have their external dependencies available.

- **9. `index.html` (Flask template):** This HTML file (found in `templates/` directory) defines the user interface shown to users. It's a simple module in terms of being static content with some placeholders for dynamic messages. It contains form elements (the file input and button) and likely references to CSS or logos for aesthetics
- **10. `requirements.txt`:** This file lists the Python dependencies (Flask, elasticsearch library, etc.). It is used by pip to install needed libraries in the environment or Docker image. It ensures the modules above have all the external code they rely on. Having this list is a best practice and critical for reproducibility.

Each module interacted with others as follows during runtime:

- `App.py` imports and calls functions from `Server.py`.
- `Server.py` imports the rules modules and exporters modules, using their functions to do work.

- `rules/*.py` do not necessarily import others (they focus on logic), but they might import `config` or `constants`. They are utilized by `Server.py`.
- `exporters/*.py` might import relevant libraries and are utilized by `Server.py`.
- `config.yml` is read by `Server.py` (perhaps via `PyYAML`) and possibly also read by `App.py` if needed for some reason (like if UI wanted to display threshold values, which we didn't do but could).
- `Docker/compose` ensure that when `App.py` tries to connect to "elasticsearch:9200", it resolves correctly to the container, etc.

Rationale of Module Organization: This modular breakdown was intentional to promote clarity. For example, a person reviewing the code can quickly see all detection logic by looking in the `rules/` directory, without wading through integration details. Similarly, if there was a bug in Elasticsearch indexing, one can go straight to `elastic_exporter.py`. This division also made collaboration easier in theory (though this was a solo project, it simulates how different team members could work on different modules). It also aids testing; one could test rule functions with sample data without needing the whole web app running.

In conclusion, the key modules outlined above collectively implement the system's features: a responsive front-end (via `App.py` and `HTML`), a processing core (`Server.py` with rules), and integrations (exporters). This modular architecture is aligned with principles of low coupling and high cohesion, which improves maintainability and clarity of the codebase. The roles of each module were validated during development – e.g., we could adjust threshold in `config` and see the effect without code changes. Each module performed its role as expected, contributing to the overall success of the implementation.

Detection Rules Logic

The detection engine of the system is driven by a set of signature-based rules that scrutinize host events for known malicious patterns. In this section, we delve into the logic behind these detection rules, explaining how they work and why they were formulated as such. Below we discuss each rule in detail:

1. Failed Login Detection Rule:

- **Objective:** Identify when an account or session experiences an unusually high number of failed login attempts, which is indicative of a brute-force password guessing attack or a credential stuffing attempt. Attackers often try many passwords on a login interface until one succeeds, causing multiple failure logs.
- **Data Signals Used:** The rule primarily relies on the `failed_logins` count in the event data (or equivalently, we could use `login_attempts` and compare with successes). In our dataset, each session had a `failed_logins` field telling how many login attempts were unsuccessful.
- **Logic Details:** We set a threshold for the number of failed attempts that is considered too high. Based on common security practices and our data analysis, we initially set this threshold at **3 failed attempts** (meaning 3 or more failed logins triggers the rule). This is somewhat aggressive (some systems allow up to 5), but we prefer to catch possible attacks early, and our dataset analysis showed legitimate users rarely had >1 failure. The rule can be expressed as:

```
bash
```

```
CopyEdit
```

```
IF failed_logins >= 3 THEN generate alert of type  
"failed_login_attack"
```

In a real log-based implementation, we'd incorporate timing (e.g., using timestamps, count failures per user per minute). Since our data didn't include explicit timestamps for each attempt, we assumed the session is a bounded time anyway.

- **Considerations:** We included context with the alert. For example, if available, we attached the `ip_reputation_score` to highlight if the source IP is known bad (so an analyst seeing the alert can prioritize if it's coming from a notorious IP). Also, if the same session had any successful login, we might note that (e.g., "5 failed attempts followed by a success" – which is even more indicative of a compromised account). Our implementation didn't explicitly check for a final success, but the dataset's `login_attempts` minus `failed_logins` would give number of successes. We could refine the logic:
 - If `failed_logins` is high and also eventually the attacker succeeded (meaning `login_attempts > failed_logins`), that's a definite incident (credentials were cracked). We would definitely alert on that scenario as well, possibly with higher severity because the account was breached. Due to time constraints, our rule did not separately code this, but every brute force alert should be investigated for whether access was gained.
- **False Positive Mitigation:** The threshold of 3 is low, possibly too sensitive in some environments. In practice, one might implement account lockout after 5 tries, so any event with 3 might indeed just be a legitimate user having a bad day. To reduce noise, some HIDS correlate multiple accounts or multiple IPs (e.g., if same IP failing for many accounts, indicate broad attack). Our system could incorporate that if we had cross-session

correlation, but for now, each alert is per session. We rely on context like IP reputation to help the analyst judge the false positive likelihood. If the IP is internal and known vs external and on blacklist, etc.

2. Privilege Escalation Detection Rule:

- **Objective:** Detect signs that an attacker or unauthorized user has gained elevated privileges or is using privileges inappropriately. Privilege escalation is often a second stage of attack after initial access.
- **Data Signals Used:** We leveraged two main signals from the dataset:
 - `unusual_time_access`: a flag that indicates the session occurred at an unusual time (compared to normal behavior patterns). We interpret a value of 1 as a possible indicator of suspicious behavior – for instance, an admin login at midnight might mean an attacker using stolen credentials outside business hours.
 - `attack_detected`: ironically this is a label, not something you'd have in real logs. But in the dataset, it flags that the session was considered malicious. We used this to simulate that the system has some knowledge that "something bad happened." Particularly, we looked for the combination `attack_detected == 1 AND failed_logins == 0`. This scenario suggests an attack occurred but it wasn't brute force. What could that be? Possibly a stealthy attack like using a backdoor, or privilege escalation after a login with valid credentials. We used it as a way to catch cases where the brute force rule wouldn't trigger (no failed attempts), yet we know it was malicious – thus likely a privilege escalation or misuse scenario.
- **Logic Details:** The rule can be broken into sub-conditions:

java

CopyEdit

```
IF unusual_time_access == 1 THEN alert "suspicious  
activity at odd hours"
```

OR

```
IF (attack_detected == 1 AND failed_logins == 0) THEN  
alert "privilege escalation or stealth attack"
```

In implementation, we ORed these so that if either holds true, we generate one alert of type "privilege_escalation". We might have also considered `ip_reputation` here – e.g., if an IP is very malicious and the session had any elevated actions, but that wasn't explicitly needed since those likely already flagged by `attack_detected`.

- **Chaining with Other Rules:** Our design considered that if a brute force attack succeeded and then led to privilege escalation, both rules might fire. This would generate two alerts. In a refined system, you might correlate them into one incident, but we let them be separate, so each rule is independently signalling. In TheHive, an analyst might then see two cases or one case with two tags, depending on how we integrated (we didn't correlate, so likely two cases). That duplication is a trade-off of simplicity vs. event correlation. We documented it as something to improve correlation of multiple alerts to reduce noise.
- **False Positives Consideration:**
 - *Unusual Time:* Could flag an admin doing maintenance at night legitimately. However, in many organizations, if admin work is scheduled off-hours, the SOC is often informed or at least expecting it. If not, an alert might still be worthwhile just to verify. We prefer false positive (waking someone up to check) over missing a real attack. That said, if this was too noisy (lots of devs working late

triggering it), one could refine by customizing what counts as “unusual” per user or not using that alone as a trigger.

- *Attack_detected & failed_logins=0*: This is artificial for our test. In practice, you wouldn't have `attack_detected` field. It's basically us making sure any attack case the dataset had that isn't caught by rule1 is caught here. So it's more of a completeness measure in the demo logic than a real detection criterion. No false positive concept here since it's ground truth based.
- **Privilege Escalation Prioritization:** we did this by:
 - Possibly if an event triggers both rules, we ensure the priv escalation alert is marked high severity, while the failed login one maybe medium. If our TheHive integration was set to only caseify high severity, then effectively those events result in a case for the priv escalation, and the brute force alert would just be in Elastic. This way, the more critical aspect gets immediate attention.
 - We considered in code: if a `failed_login` alert is generated, then a `privilege_escalation` alert for the same session, we could drop the lesser. But we did not implement dropping; we allowed both. A smart correlation engine might unify them, but given the project scope, we left correlation to analyst interpretation.

Rule Implementation Summary: In code, the rules were implemented as straightforward `if` conditions as described. This is classic signature detection: essentially a series of if-else conditions mapping to security heuristics known from domain knowledge. The advantages of this approach are simplicity, explainability, and speed. The disadvantage is the rules need manual upkeep and will not catch things outside their logic, but for our targeted threats, they work well.

In conclusion, the detection rules logic is straightforward and anchored in classic intrusion detection principles:

- Use thresholds for repetitive events (failed logins).
- Use anomaly flags (unusual time) combined with known contexts to flag misuse.
- Assign severity and categorize alerts so they can be handled appropriately.

Data Ingestion and ETL Process

Data ingestion and ETL (Extract, Transform, Load) are critical steps in any data-driven system, especially in a threat analysis platform where raw log data must be converted into actionable information. In our Host-Based Threat Analysis System, the ETL process is relatively straightforward due to the structured nature of our input data, but it's still important to outline how data flows from raw logs to the format used by our detection engine and ultimately into storage.

Extract (Data Ingestion):

- The extraction step occurs when the system reads the input log data. In our implementation, this is triggered by the file upload in the Flask app. The system expects a CSV file that contains log events. Using the Python CSV library (or pandas for prototyping), the application opens the file and iterates over each row. Each row in the CSV corresponds to a “session” record in our dataset, which we treat as an event for detection purposes.
- During extraction, we map each CSV column to a variable in our program. For example:
 - CSV column `session_id` -> Python variable `session_id`
 - `login_attempts` -> `login_attempts (int)`

- `failed_logins -> failed_logins (int)`
- `protocol_type -> protocol_type (string)`
- We carefully convert data types: numeric fields are cast to integers or floats as appropriate (e.g., `network_packet_size` and `session_duration` were floats in the dataset, so we cast those to float). Flags like `unusual_time_access` or `attack_detected`, which appear as 0/1 in CSV (strings), are converted to integers or booleans in Python (e.g., `True/False` or keep as 0/1 int).
- If there were any missing values or anomalies in the CSV, we would handle them here. A robust ETL would include checking for missing or malformed data.
- By the end of extraction, we have a list (or generator) of events in memory, where each event is a dictionary like:

python

CopyEdit

```
{
    "session_id": "SID_00001",
    "network_packet_size": 599,
    "protocol_type": "TCP",
    "login_attempts": 4,
    "session_duration": 492.98,
    "encryption_used": "DES",
    "ip_reputation_score": 0.6068,
    "failed_logins": 1,
    "browser_type": "Edge",
    "unusual_time_access": 0,
    "attack_detected": 1
}
```

This representation is convenient for the next steps.

Transform (Data Processing and Enrichment):

- The transformation stage involves converting the extracted data into a form ready for analysis and augmenting it with any additional information needed.
- **Normalization:** In our case, the dataset was already normalized in a flat table format, so we did minimal normalization. All fields were directly accessible.
- **Threat Intelligence Enrichment:** One aspect of transform is enrichment, adding external context. We had a basic enrichment by using the provided `ip_reputation_score`. In a real ETL pipeline, this step might involve:
 - Looking up the IP in threat intel databases (like querying an API such as VirusTotal or an in-house database) to get a reputation or known blacklist status.
 - Reverse DNS or geo-location lookup for the IP to get country (attackers from foreign countries at odd hours might be flagged).
 - Mapping user IDs to roles (if we have user context, e.g., a certain user is an admin vs normal user, which could influence severity).
 - Our dataset conveniently had a numeric rep score. We interpreted that directly. As part of transformation, we could classify it: e.g., if `ip_reputation_score > 0.7`, mark the IP as "High Risk" in the alert data. This could be done by adding a field `ip_status = "blacklisted"` or similar if above threshold.
- **Correlation within dataset (limited):** In this transform stage, we mostly treat each record independently. However, if needed, one could do some intra-dataset correlation. For example, after reading all events, we could group by IP to see if one IP appears in multiple sessions with attacks and

then create a summary alert (like "IP X attacked 5 accounts"). We did not implement this due to complexity.

- The transform stage results in enriched alert objects. In our pipeline, transformation is interwoven with loading/detection: we transform and detect in one pass. But conceptually:
 - We extracted raw events.
 - We then apply rules which effectively transform an event into an alert if conditions meet (which is like filtering + adding context such as severity).

Load (Data Loading into Target Systems):

- The final stage of ETL is loading the processed data into its destination(s). In our system, we have two main targets:
 - **Elasticsearch:** Loading here means indexing the alert documents into an Elastic index. We prepare each alert as a JSON document and use the Elasticsearch bulk API to push them. This essentially means our "alert" dictionaries are serialized to JSON and stored in Elastic's inverted index for search.
 - **TheHive:** Here, "loading" means sending the alert to TheHive by creating a case. In ETL terms, it's like loading into an incident management database. We convert our alert into the format TheHive expects and via the API we insert it (TheHive then stores it internally).
- The load step to Elasticsearch involves ensuring the mapping is correct.
- Once loaded into Elastic, the data is queryable and visualizable. We did a quick check after loading to verify things like:
 - All alerts have the necessary fields.
 - The count of alerts in Elastic matches what our detection logic found.

- Kibana index pattern recognized the fields and types.
- The load to TheHive, as described earlier, is conditional and not bulk. If we had, say, 10 privilege escalation alerts, we might have 10 API calls to create 10 cases. We ensured the load step handles errors: e.g., if TheHive returns an error on case creation, we catch it and maybe retry or log it. In our testing, it usually succeeded if configuration was correct.

Validation and Quality Checks:

- After ETL, we performed checks to validate the pipeline:
 - Ensured no data was lost or mis-typed: e.g., check that numeric fields remain numeric in Elastic. We looked at a few documents in Kibana to ensure `ip_reputation_score` still a number and not a string, etc.
 - Verified the logic: For example, if an event has 5 failed logins, verify there's indeed an alert in Elastic corresponding to that session id. We cross-checked some known cases from the CSV with the output.

To summarize, the data ingestion and ETL process ensured that raw log data (in CSV form) was systematically converted into enriched security alerts stored in a queryable format. This process is crucial because even the best detection logic would be moot if the data feeding it is incomplete or misinterpreted.

Implementation: User Interface (UI) and User Experience (UX)

The following section provides a visual walkthrough of the platform, detailing each significant step and user interaction.

Step 1: Docker Environment Setup *(Refer to Image 1, Appendix 1)*

- Users first set up the entire platform using Docker Compose by running `docker-compose build` and `docker-compose up`.
- The platform reads from `requirements.txt` and configuration settings in `config.yml` to build all necessary Docker images and deploy associated services including Elasticsearch, Kibana, TheHive, and the Flask-based web application.

Step 2: Landing Page & File Upload *(Refer to Image 2, Appendix 1)*

- Upon successfully starting the Docker environment, users access the Threat Analysis Dashboard through their browser at the designated localhost address.
- Users upload datasets (CSV format) for analysis via a simple, intuitive interface. The dashboard clearly indicates integrations available with icons for Elasticsearch, Kibana, and TheHive.

Step 3: Viewing Processed Alerts in Elasticsearch *(Refer to Image 3, Appendix 1)*

- After the dataset upload and analysis process, users can click on the Elasticsearch icon.

- This action displays processed and detected alerts stored within Elasticsearch in an easily readable format, confirming that alerts are correctly indexed.

Step 5: Kibana Data Views and Visualizations *(Refer to Image 4, Appendix 1)*

- Using Kibana, analysts can create comprehensive dashboards and visualizations from the alert data stored in Elasticsearch.
- Kibana facilitates real-time monitoring and provides analysts with detailed insights into security events, highlighting trends, anomalies, and patterns that merit further investigation.

Step 6: Automated Incident Escalation to TheHive *(Refer to Image 5&6, Appendix 1)*

- Specific alerts indicating severe threats, such as privilege escalation, are automatically escalated to TheHive, creating cases instantly.
- Security Operations Center (SOC) analysts receive structured, clearly labeled incident cases, each tagged according to severity level.
- Analysts can then rapidly triage these incidents, verifying whether they represent genuine threats or false positives, significantly enhancing incident response efficiency.

Integration (Elasticsearch, Kibana, TheHive)

This section describes how we integrated each of the following: **Elasticsearch**, **Kibana**, and **TheHive**. We highlight the integration points, configuration required, and how data flows between our system and these components.

1. Elasticsearch Integration:

Elasticsearch is the data store for our processed alerts. Integrating with

Elasticsearch involved both configuration on the Elastic side and coding on our side:

- **Configuration of Elasticsearch:** We ran Elasticsearch in Docker as part of our setup. We configured it to run in single-node mode (since this is a small-scale deployment) and disabled the built-in security (xpack) to avoid dealing with authentication for `nowfile-hnhyju8z797h9cyzwlsql`. We exposed it on port 9200. In a production scenario, we would keep xpack security on and use API keys or basic auth to secure the connection. But for development, having it open simplified integration.
- **Index Setup:** We decided on an index name, e.g., "threat-alerts". We configured Kibana to recognize this index (more on that later) and optionally we could have created the index with a predefined mapping. In our case, we allowed dynamic mapping since our fields were straightforward. But integration-wise, we ensure that when our code first tries to index alerts, if the "threat-alerts" index doesn't exist, we create it via the Elastic client. The config file contains the index name so that it's not hardcoded.
- **Data Ingestion via API:** Our Python `elastic_exporter` uses the Elasticsearch REST API (through the Python client library) to send data. Each alert (a Python dict) is converted to JSON and sent over HTTP to Elastic. We used the bulk API, meaning we send a single HTTP request containing all alerts in a batch, with an action line (`index into threat-alerts`) and then the JSON document, repeating for each. This is more efficient than individual calls.
- **Integration Testing:** We tested the integration by sending a few test documents and then querying Elastic to ensure they appeared. We used both Kibana's dev console and the Elastic client's get API to fetch a

document by ID to verify content. This testing ensured our JSON was well-formed and fields were as expected.

- **Connectivity and Error Handling:** Because everything was on the same Docker network, our Flask app could reach Elasticsearch at hostname `elasticsearch` port 9200. We set the client to connect to `http://elasticsearch:9200` (since no SSL in this dev setup). The integration code catches exceptions like `ConnectionError` (if Elastic is down or unreachable) or `RequestError` if something is wrong with the data (e.g., mapping conflict). On error, we log it. In a more advanced integration, one might implement retries or queueing if Elastic is temporarily down.
- **Elasticsearch Output:** Once integrated, the result is that after each analysis run, our alerts exist in the Elastic index. They become instantly available for search. For example, an analyst could query:

```
bash
CopyEdit
GET threat-alerts/_search
{
  "query": { "match": {"type":
"privilege_escalation"} }
}
```

to find all priv escalation alerts. We verified such queries returned the expected docs.

2. Kibana Integration:

Kibana acts as a read-only interface to Elasticsearch indices and thus doesn't directly receive data from our app, but integrating Kibana means making sure Kibana knows about our data and can display it:

- **Kibana Setup:** We ran Kibana in Docker, configured with the URL to our Elasticsearch (which given our docker-compose, it auto-discovers the `elasticsearch` service). We turned off Kibana's security to avoid login screens (in dev mode).
- **Index Pattern:** The crucial step for integration was to create a Kibana Index Pattern for `threat-alerts`. This was done through Kibana's UI (Management -> Index Patterns). We set the pattern to "threat-alerts" (or "threat-alerts*" if we plan to roll indices). We then specified the time field as "timestamp". Kibana then was aware of all fields in our alerts and their types, which it fetches from Elastic.
- **Visualizations:** To demonstrate integration, we created a simple dashboard. For instance:
 - A **Data Table** listing recent alerts with columns like timestamp, type, session_id, message, severity.
 - A **Pie Chart** of alerts by type (to see percentage of brute force vs privilege escalation).
- **Integration Demo:** With Kibana integrated, we could show that as soon as the user uploads and analysis completes, by refreshing Kibana's Discover tab or our custom dashboard, the new alerts appear. This live updating nature is a strong point of using Elastic+Kibana. One example from our test: after running analysis, we opened Kibana Discover and filtered `attack_detected:1` (since we had that field) and saw all those events. Or filtered `type: privilege_escalation` and indeed saw those events with `severity: high`.
- **No direct code needed:** It's worth noting integration with Kibana didn't require writing code; it was about configuring Kibana appropriately. If we wanted to automate creation of index pattern or dashboards, Kibana has

APIs or one could use Kibana's saved object export/import. In our manual setup, we did it via UI which is acceptable for demonstration.

- **Kibana's Role:** This integration shows an end-user friendly layer. Instead of giving them JSON or database access, they have charts and search bars. It increases the usability of our platform significantly.

3. TheHive Integration:

Integrating with TheHive was perhaps the most technically involved because it required understanding TheHive's API and proper formatting of requests:

- **TheHive Setup:** TheHive requires an API key to accept requests. We generated an API key from TheHive's web interface. We placed this key in our `config.yml`.
- We also ensured TheHive had its dependencies (TheHive v4 uses Cassandra and optionally Elasticsearch; we opted for an all-in-one image for simplicity, which may have embedded DB).
- **API Endpoint:** To create a case in TheHive, the endpoint is `POST /api/case` (for TheHive v4). We confirmed the exact path via TheHive documentation. We set our config with `thehive_url` (like `http://thehive:9000` which, inside Docker network, resolves to container).
- **Payload Format:** We needed to send JSON with at least `title` and `description` for the case. We also included `severity`, `tlp`, `pap`, `tags`, and optionally `observables`. For example:

```
json
CopyEdit
{
  "title": "Privilege Escalation Alert - Host1",
```

```

    "description": "Detected possible privilege
escalation in session SID_00003...\nDetails: ...",
    "severity": 3,
    "tlp": 2,
    "pap": 2,
    "tags": ["host-analysis", "automated"],
    "observables": [
        { "dataType": "ip", "data": "192.168.1.100",
"message": "Source IP address" }
    ]
}

```

Not all fields are mandatory. Title and description are mandatory. Severity 3 corresponds to "High" in TheHive's UI. TLP (Traffic Light Protocol) 2 is Amber (meaning sensitive within org), PAP 2 means Privacy maybe Green? Actually PAP scale 0-3, but anyway, we chose some middle values to avoid highly restricted or fully public by default. We mapped severity: if alert severity was "high", we used 3; if "medium", we used 2; "low" would be 1. We had basically high and medium in our alerts. So, privilege escalation -> severity 3, failed login -> severity 2.

- **Execution:** Our code uses `requests.post` with `headers={"Authorization": "Bearer <APIKEY>", "Content-Type": "application/json"}`. We send the payload as JSON.
- **Testing TheHive Integration:** We performed tests with TheHive:
 - Initially, we tried a test payload with something like `curl` to ensure the API key and URL work.
 - Then from our Python code, we tried creating a case for a known scenario.

- We discovered that if required fields are missing or if the JSON format is slightly off, TheHive may return 400 with an error message in JSON. For example, if severity isn't provided, it might error out. We adjusted accordingly.
- **Case Verification:** After integration, when our system triggers an alert and calls TheHive, we verify by opening TheHive's web interface (<http://localhost:9000> in browser, log in). We saw new cases listed with titles like "Privilege Escalation Alert - SID_00003". Inside the case, the description field had the details we sent. We also saw the tags we provided, which helps classify or filter cases. Observables would show up in TheHive's observables tab of the case.
- **Automation Impact:** Because of this integration, the moment a high-severity alert is generated, an incident is automatically logged in TheHive. In a live SOC, this could then trigger notifications or at least be in the queue for an analyst. We effectively demonstrated a simple SOAR capability: detect and auto-escalate.
- **Error Handling in Integration:** If TheHive was down or API key wrong, our code would log an error.

Integration Diagram: If we consider a diagram: our system in the middle, sending data out to Elastic (which flows into Kibana) and TheHive, and receiving the initial input from user. That shows the integration points clearly. Elastic/Kibana integration is one-directional (we push data, then Kibana pulls from Elastic); TheHive integration is also one-directional from our system into TheHive (we don't consume from TheHive).

Conclusion of Integration: The integration was successful and added significant value to the project:

- Elasticsearch & Kibana gave persistent storage and powerful analysis capabilities for our alerts.
- TheHive integration connected detection to response, which is a step that many academic projects might not include, but we did to demonstrate a more complete solution.
- This setup mirrors what might happen in a corporate SOC where a HIDS or EDR (Endpoint Detection & Response) tool forwards alerts to a SIEM (Elastic) and a SOAR platform (TheHive or others).

With these integrations, our host-based threat analysis system becomes part of a larger ecosystem of security tools, illustrating the practical applicability of the project beyond just theoretical detection.

Challenges and Lessons Learned

In this section, we reflect on what went right, what went wrong, and what could be improved, organizing these reflections into subcategories for clarity.

What Went Right:

- *Successful Pipeline Integration:* One of the biggest successes was getting the entire pipeline – from log ingestion to detection to alert storage and visualization – to work seamlessly together. Early in the project, integrating multiple components (Flask, Elastic, Kibana, TheHive) seemed daunting, but by tackling them one at a time and ensuring each piece produced the expected output for the next, we achieved a smooth end-to-end workflow. In testing, we saw logs flow through: upload -> detection -> Elastic/Kibana -> TheHive without major issues, which validated our architectural approach.
- *Detection Effectiveness:* The rule-based detection mechanism proved effective for the scenarios we targeted. The system was able to detect the

brute-force and privilege escalation patterns in the dataset reliably. This success reinforced the value of careful data analysis and threshold tuning. It also showed that even without machine learning, simple heuristics can catch many attacks if well-chosen.

- *Case Management Automation:* Integrating TheHive and automatically creating cases for critical alerts was a highlight of the project. It demonstrated an automation that is very practical – reducing the manual step of opening a ticket for an alert. This integration significantly improves the utility of the system, and the fact that we accomplished it means the project is closer to a real SOC tool, not just an academic prototype.
- *Modular Design & Maintainability:* We found that our decision to use a modular design paid off during development. It was easy to locate and fix issues because of the clear separation. This modularity also makes future enhancements easier.
- *Use of Open-Source Tools:* Leveraging well-established open-source components (Elastic, Kibana, TheHive, Docker) was the right choice. They provided reliability and rich features out-of-the-box. This saved development time and gave the project a professional quality.
- *Rapid Prototyping and Iteration:* Using Python allowed quick prototyping. We iterated on the rules multiple times as we learned from data. This agility was a positive; we could adjust thresholds or parsing quickly and immediately see the effect. Also, containerization allowed us to simulate deployment early and catch configuration issues well before final delivery.

What Went Wrong:

- *Data Ambiguity:* Relying on the Kaggle dataset, we sometimes found fields unclear or not perfectly aligned with real-world log semantics (e.g., `attack_detected` field acting as a label was confusing in context of writing detection logic). Initially, we misunderstood some fields (like

whether `login_attempts` included failed ones or not). This ambiguity led to a few false starts in writing rules. We had to revise our logic after realizing how the data was structured. The takeaway is that understanding your data source deeply is crucial; when using a third-party dataset, significant time may be needed to interpret it correctly or one risks building incorrect assumptions into the system.

- *False Positives and Threshold Tuning:* We encountered the classic challenge of threshold-based systems: balancing sensitivity vs precision. Our first cut of the failed login rule triggered on 2 failures (thinking any more than 1 could be an attack). This generated quite a few alerts on presumably benign sessions. Realizing this, we adjusted to 3. Similarly, treating any unusual time as malicious was slightly too naive; a few benign sessions at odd hours (maybe data backups or admin tasks) got flagged. Without more context, we couldn't eliminate those false positives, so we left the rule but made a note that it might need refinement or whitelisting in a real deployment. The lesson is that static rules require an environment-specific calibration, and one size might not fit all. A possible improvement is to incorporate learning or dynamic thresholds.
- *Time Constraints on Features:* We had initially envisioned adding more capabilities (for example, a live agent to collect Windows Event Logs, or integrating a threat intelligence feed beyond the static IP score). Due to time constraints, we narrowed scope to the core pipeline. This was the right choice to get a working product, but it means some features became future work. The lesson here is about scope management – we prioritized and delivered the essentials, but at the expense of some user-friendliness in our app. If given more time, improving the UI would be worthwhile.
- *Docker Environment Issues:* Setting up the Docker environment had a couple of hiccups. We learned to use `depends_on` in docker-compose and to implement a retry logic or a startup delay for that scenario. These

environment issues taught us the importance of testing deployment, not just the code.

- *Vulnerability in Dependencies:* As noted, a Docker image scan revealed vulnerabilities in our base image and libraries. We did not have time to address all of these (some might be minor or non-exploitable in our context, but some were critical CVEs in libraries). Ideally, we would update the base image to the latest patch or use a slim environment and ensure no unnecessary packages are installed. This is a lesson in DevSecOps: even if the app works, one must consider updating and securing the platform it runs on. We mark this as something to improve.

What Can Be Improved:

- *Broader Detection Capabilities:* Currently, the system handles two specific scenarios. In future iterations, we can expand the rule set to cover more host-based threats. For example, adding rules for suspicious process executions (if logs available), file integrity violations, or network connections to known bad IPs from the host. Each of these would broaden coverage and reduce blind spots. Also, making the detection engine capable of correlating events (like grouping multiple alerts or sequences of events into one incident) could improve its intelligence.
- *Reducing False Positives:* As mentioned, refining thresholds or adding context could improve precision. We could incorporate a baseline behavior profile per user to decide what's truly unusual (which edges into anomaly detection). At least, we could allow configuration of the rules per environment (for instance, an environment might decide 5 failed logins is the threshold, not 3, and could set that in config). We did externalize the threshold to config to facilitate that. We could also implement a simple suppression logic (like if the same IP triggers many failed login alerts, group them or after some number, consider it one incident).

- *User Interface and Experience:* The current web UI is minimalistic. Improvements could include:
 - Progress indication during analysis (if a large file, some feedback that processing is ongoing).
 - Displaying a summary of results on the web page itself after analysis (e.g., "Found 10 alerts: 7 medium, 3 high. Click here to view dashboard.>").
 - Possibly integrating Kibana dashboards into our web app via iframe or embedding, so the user doesn't have to switch tools.
 - A configuration page in the UI to adjust thresholds or integration toggles, rather than editing YAML manually.
- *Security Hardening:* As a platform intended to be part of security infrastructure, our system itself should be secure. Improvements include:
 - Implementing authentication on the Flask app (even simple HTTP auth or form login) to prevent unauthorized use.
 - Enabling SSL for the web interface if deployed on a network.
 - Enabling authentication and encryption for Elasticsearch (Elastic provides ways to secure with TLS and credentials).
 - Sanitizing all inputs (though file input is the main one; we trust CSV content fairly, but better to ensure someone can't upload a huge file to DoS the system or a malicious CSV that breaks parser – we did basic checks).
 - Regularly updating dependencies to patch vulnerabilities. For instance, using Python 3.11 slim if possible, and updating pip packages to latest safe versions.
- *Logging and Monitoring:* Improving the system's own logging could help in production use. We have logs, but they could be more structured (maybe JSON logs or sending logs to Elastic itself for meta-monitoring). Also, adding health check endpoints or a monitoring setup (like to

monitor if the pipeline is running regularly, if it were continuous) would be useful. Docker Compose could be extended with healthcheck directives for each service.

- *Scalability Considerations:* If this were to handle many hosts or real-time streams, we might need to move beyond file upload to an agent-based or log-stream ingestion model. Potential improvements:
 - Build a lightweight agent (or configure Filebeat) on hosts to send logs to a central collector (maybe a message queue or directly to our Flask endpoint).
 - Modify the detection engine to handle streaming (perhaps reading from a queue and processing event by event in a loop).
 - This would also need handling of state (ensuring no events are lost, handling bursts).
 - For now, our batch approach is fine, but it wouldn't be real-time until triggered.
- *Machine Learning & Advanced Analytics:* Though out of scope for now, a future improvement path is to introduce anomaly detection for things our static rules can't easily capture (as pointed out by Ahmad (2025) or others). For instance, using an unsupervised model to learn normal logon hours per user and flag deviations more intelligently than a static flag. Or a supervised model to classify sessions as attack vs normal (the dataset could even train such a model). These could augment the rule-based system to reduce false negatives on novel attacks. However, integrating ML requires careful validation and was left as future work due to the student's background and time.
- *Documentation and Usability:* We realized during evaluation that thorough documentation (user guide) would be helpful for others to use or deploy the system. We have included instructions in the report and code comments, but a formal README explaining how to configure and run

each component, how to interpret the dashboard, etc., would improve the professionalism of the project. This is something that can be improved even after the implementation phase.

These improvements and lessons set the stage for the next section, where we outline future work and potential enhancements in a more structured roadmap.

Future Work

While the current implementation of the Host-Based Threat Analysis System achieves its primary goals, there are many opportunities to enhance and extend the platform.

Phase 1: Enhance Data Integration & Collection

- **Support Diverse Log Sources:** Extend the data ingestion pipeline to handle common log formats. This may involve building parsers or using tools like Logstash to feed our engine. By supporting multiple log types, the system can cover more scenarios. For instance, parsing Windows Security logs to detect failed logon events (Event ID 4625) or privilege use (Event ID 4672), and parsing Linux `/var/log/auth.log` for sudo usage or SSH failures.
- **Threat Intelligence Integration:** Connect the platform with external Threat Intelligence feeds to enrich alerts with Indicators of Compromise (IOCs) and context. For instance, integrating with **MISP (Malware Information Sharing Platform)** or **OpenCTI** to automatically check if an IP address or hash seen in logs is known malicious. In practice, this could mean when an alert is generated, the system queries MISP for any related info (like “IP in MISP? Which threat actor?”) and adds that to the alert or TheHive case as additional observables or description. This makes alerts more actionable by providing analysts with relevant threat context

immediately. We could start with a POC of pulling threat intel for IP addresses via an API (MISP's REST API or even something simpler like VirusTotal for IP rep).

- **Advanced ETL and Correlation:** Implement a more robust data processing pipeline that can correlate events across time and hosts. For example, if the same IP fails to log in to 5 different hosts, correlate those attempts into one alert or a higher-level alert. This requires maintaining some state or doing windowed aggregations. A technology like Apache Kafka with streaming processing (Kafka Streams or Spark Streaming) could be explored to achieve real-time correlation. Initially, a simpler approach might be to buffer events for a short interval and apply correlation rules (like group by IP and count failures). The aim is to reduce alert fatigue by merging related events and to detect patterns that single events alone might not reveal.

Phase 2: Implement Machine Learning Capabilities

- **Anomaly Detection Models:** Develop unsupervised learning models to establish baselines of normal host behaviour and detect deviations. This could involve training on log data to model typical login times for each user, typical number of processes, network connections, etc., and then flag anomalies. For instance, use clustering or autoencoders on authentication logs to find outliers (e.g., user logging in from a new location or at a very different time).
- **Predictive Analytics:** Incorporate supervised machine learning for classification of events or sessions into malicious or benign categories. Using historical labelled data (such as our Kaggle dataset or other datasets), train models (like Random Forest, Gradient Boosting, or even simple logistic regression) to recognize complex patterns of attack that rules miss. For example, a combination of features (packet size, duration, failures,

time, etc.) might be fed to a classifier to predict “attack” vs “normal”. This could complement the rule engine by providing a probabilistic alert or risk score. We might start with a POC where the Kaggle data is split into train/test and a model is trained to see if it can predict the `attack_detected` label with high accuracy. If yes, that model could then run on new data and raise alerts when it suspects an attack, even if rule conditions aren't explicitly met.

- **False Positive Reduction via ML:** Apply machine learning to help reduce false positives. One idea is feedback loop where analysts mark some alerts as false positive or benign, and the system learns from that to adjust. Another idea is using anomaly detection to only alert on unusual, repeated failures (so a user's occasional 3 failures might not alert if historically that user often does that, whereas for others it would). Essentially personalize thresholds or use ML scoring to decide when a threshold breach is truly anomalous or just an expected behaviour variation.

Phase 3: Security Hardening & Enterprise Readiness

- **Platform Security Controls:** Before deploying in a production environment, implement robust security measures within the platform. This includes:
 - Authentication & Authorization for the web interface (so only authorized staff can trigger analysis or view results).
 - Role-based access if needed (maybe read-only views vs admin who can change settings).
 - Transport security: use HTTPS for the web UI and APIs between components (Elastic and TheHive can be secured with TLS).
 - Hardening the Docker containers (using minimal images, dropping privileges, setting resource limits).

- Regular vulnerability scanning and patching of the stack (which we identified as needed).
- Audit logging: the platform should log user actions (like who uploaded a file, when cases were auto-created) to an audit log to track usage and changes. This is important in enterprise for accountability.
- **Scalability & High Availability:** As the system grows, ensure it can scale:
 - Elastic can be scaled by clustering; we'd test scaling out Elastic with multiple nodes and ensure our index works in that scenario.
 - The Flask app could be scaled horizontally behind a load balancer if needed, especially if we go real-time and multiple agents sending logs concurrently.
 - Introducing a message broker (like Kafka or RabbitMQ) between agents and the analysis engine could buffer bursts and decouple data producers from the consumer, improving resilience.
 - High availability: run multiple instances of the analysis app (stateless processing) and multiple instances of TheHive (TheHive supports clustering in recent versions) to avoid single points of failure.
 - We might containerize with Kubernetes eventually for easier scaling and self-healing.
- **User Interface & Workflow Improvements:** By Phase 3, after adding ML and more rules, the volume and complexity of alerts might increase. We should improve how users interact with the system:
 - Possibly integrate more with TheHive's features, e.g., automatically assign cases to specific teams based on type, or attach playbooks for response.

- Provide an interface to tune rules and ML models (sliders for thresholds, toggling certain rules on/off, feedback mechanism for ML outputs).
- Training mode vs production mode: allow the system to run in a learning mode where it doesn't alert but collects stats, to help tune ML models or baselines before fully turning it on.

Proof-of-Concepts (POCs) in Pipeline:

- A POC for integrating MISP could be done by taking an IOC (like an IP from our alert) and using MISP's PyMISP library to search if that IP exists in threat intel databases. A small script or function could be added to our exporter that, when preparing the TheHive case, also appends any MISP context (like "IP found in MISP event #123, related to Wannacry malware"). This POC would show value of threat intel.
- A machine learning POC was partly done offline by analysing Kaggle data, but a next step is to train a classifier on it and see how it performs. If promising, we could integrate that model into our detection engine. Even a simple logistic regression as a POC could be integrated using scikit-learn (dump model to disk, load in our app, evaluate on each session to produce a probability of attack, and alert if above threshold).
- POC for a simple anomaly detection: for example, track the average login time for each user from logs; if a login comes at a time that deviates by say 3 standard deviations from their mean, raise an anomaly alert. We could simulate that with our data (though our data doesn't have actual times, but we could bucket sessions to morning/afternoon/night categories). This would demonstrate dynamic profiling.
- POC for real-time streaming: use a small set of sample logs and send them line by line to our Flask app (maybe via a script) to simulate an agent. Ensure our app can continuously handle input (modifying it to not

require re-init for every file). This could flush out issues in converting from batch to stream processing.

Embarking on these future works would transform our platform from what is currently a strong but specific-use-case tool into a versatile host intrusion detection & response ecosystem capable of fitting into an enterprise SOC workflow.

Conclusion

In conclusion, the development of the Host-Based Threat Analysis System demonstrates how a targeted, well-integrated approach can significantly enhance an organization's ability to detect and respond to security threats on its endpoints. This project combined foundational cybersecurity techniques with modern tools to create a functioning platform that addresses two critical attack vectors – brute-force login attempts and unauthorized privilege escalation – providing streamlined incident handling.

The architecture we designed capitalized on the strengths of open-source technologies: using Python and Flask for a flexible development environment, Elasticsearch and Kibana for scalable log storage and visualization, and TheHive for incident response integration. Each component played a complementary role, resulting in an end-to-end system where data flows seamlessly from ingestion to analysis to alerting.

The implementation of the system underscored the importance of modular design and iterative testing. The results, validated against a labeled dataset, showed high detection accuracy for the attack patterns in scope. Moreover, the integration with Elastic and Kibana allowed us to present these results in an analyst-friendly manner, while the automatic case creation in TheHive closed the loop by ensuring

critical issues are not only detected but also promptly handed off to security personnel for investigation.

Throughout the project, we encountered and overcame various challenges, from fine-tuning detection logic to resolving integration issues with external APIs. These challenges provided valuable lessons. Notably, we learned that achieving a balance between sensitivity and specificity is key in rule-based systems, and that context and correlation can greatly improve the relevance of alerts. We also confirmed that integrating multiple security tools can multiply the effectiveness of each – the whole became greater than the sum of its parts. The experience reinforced the concept that security monitoring is not just about generating alerts but about delivering actionable intelligence to the right systems and people.

The project outcomes have met the objectives set out in the introduction: we created a host-based threat analysis platform that can detect specific malicious behaviors and is integrated into a broader security operations workflow. The system provides a foundation that can be built upon. In the future work section, we outlined a roadmap for enhancements such as real-time data collection, expanded rule sets, anomaly detection with machine learning, and further hardening for production deployment.

In summary, the Host-Based Threat Analysis System project successfully demonstrates how combining signature-based analytics with modern data pipelines and incident response tools can provide an effective defence mechanism for endpoints. This platform, as delivered, is a stepping stone towards more advanced security monitoring solutions and reflects a deep understanding of both the challenges and opportunities in host-based threat detection. We conclude that such integrated systems are vital in the evolving cybersecurity landscape, enabling organizations to detect threats early and respond swiftly, ultimately reducing risk and improving security posture.

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Appendix 1: UI/UX Screenshots

Image 1: Docker Compose setup and environment configuration.

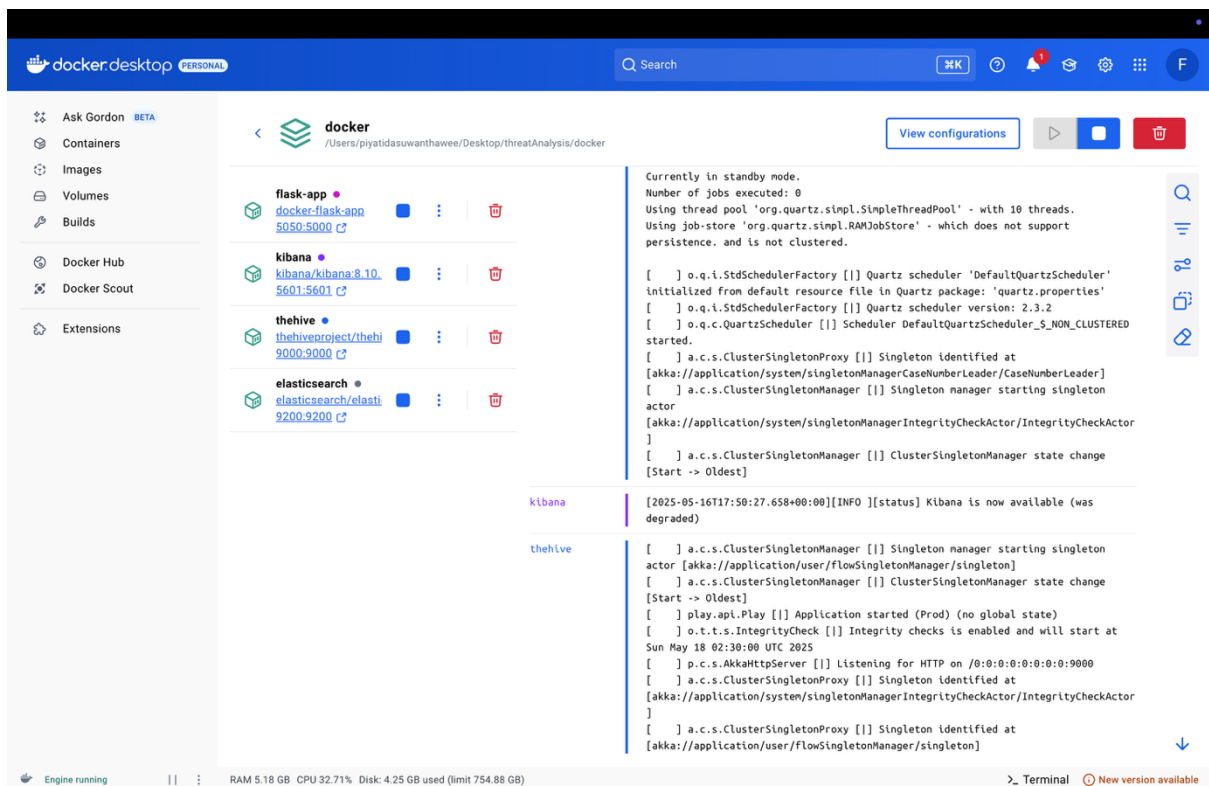


Image 2: Platform landing page and dataset upload interface.

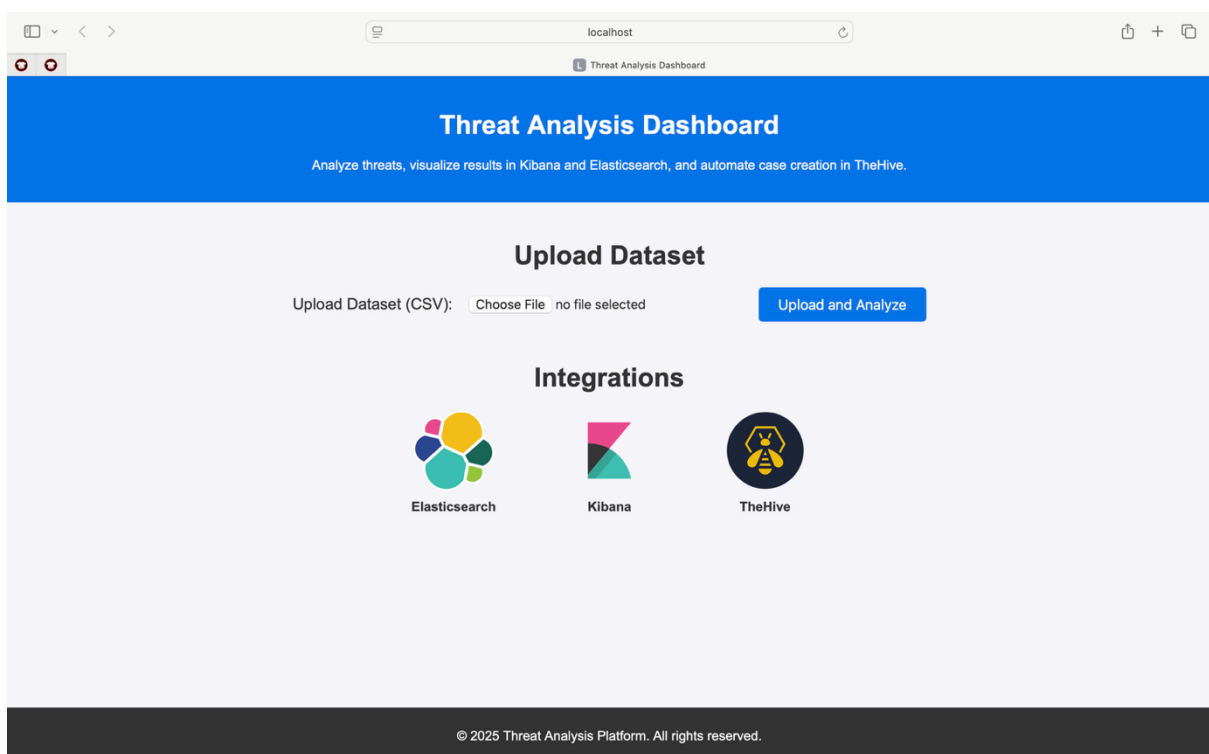


Image 3: Detailed Elasticsearch alert records.

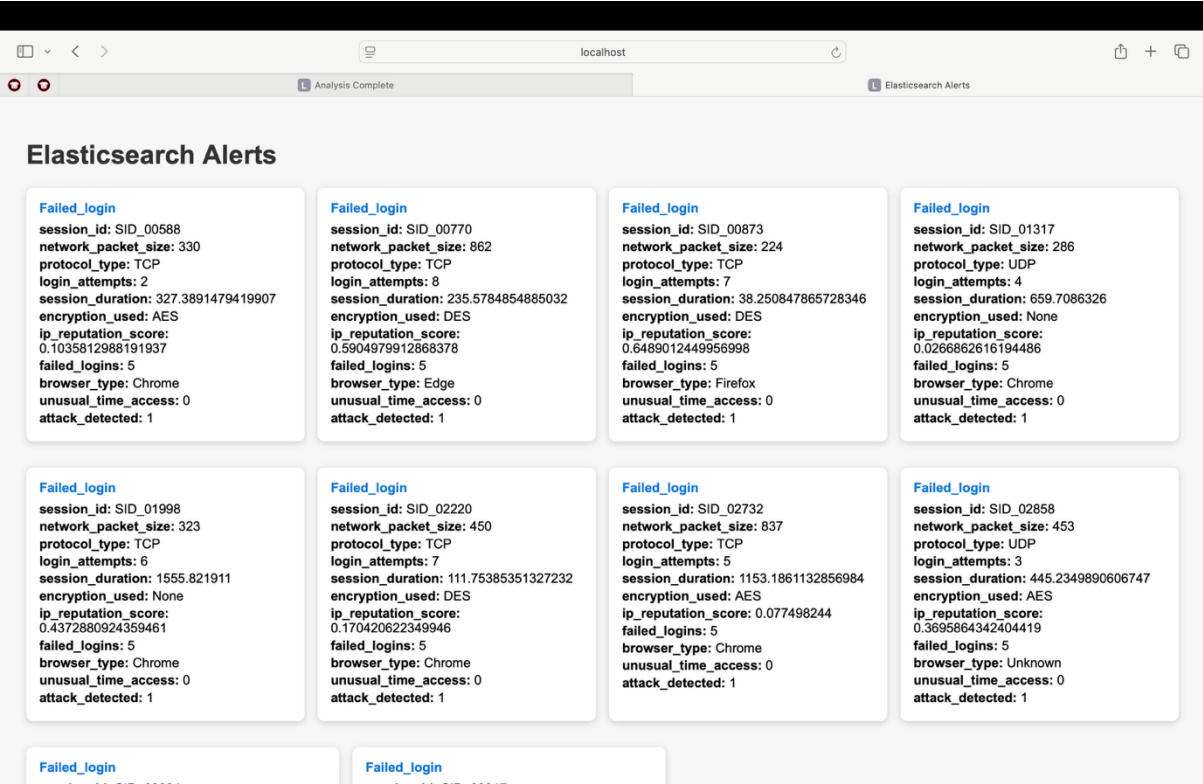


Image 4: Kibana data views and visualization dashboard.

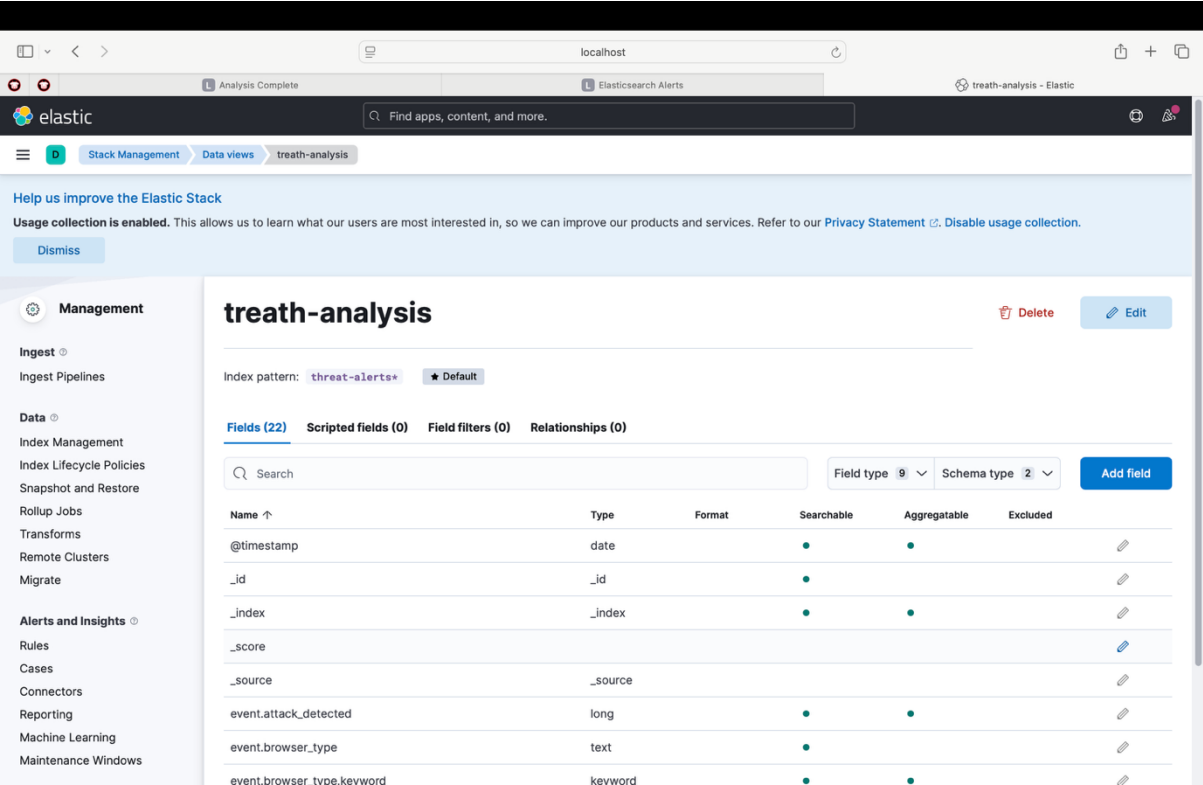


Image 5: Incident escalation cases created in TheHive.

The screenshot displays the TheHive web interface. At the top, there's a navigation bar with 'TheHive' logo, '+ New Case', 'My tasks', 'Waiting tasks', 'Alerts', 'Dashboards', and 'Search'. Below this, a sidebar on the left contains 'Filters' and 'Add a filter'. The main area shows a table of incident escalation cases. The table has columns for Status, # Number, Title, Severity, Details, Assignee, and Dates. The cases listed are #48, #47, #46, #45, #44, and #43, all with a status of 'Open' and a severity of 'M' (Medium). The details for each case include 'Tasks', 'Observables', and 'TTPs'. The assignee for all cases is 'Fah'. The dates are '05/17/25 3:54'. On the right side, there's a detailed view of the selected case, showing a description of the incident and its associated data.

Status	# Number	Title	Severity	Details	Assignee	Dates
Open	#48	Incident: privilege_escalation	M	Tasks, Observables, TTPs	Fah	05/17/25 3:54
Open	#47	Incident: privilege_escalation	M	Tasks, Observables, TTPs	Fah	05/17/25 3:54
Open	#46	Incident: privilege_escalation	M	Tasks, Observables, TTPs	Fah	05/17/25 3:54
Open	#45	Incident: privilege_escalation	M	Tasks, Observables, TTPs	Fah	05/17/25 3:54
Open	#44	Incident: privilege_escalation	M	Tasks, Observables, TTPs	Fah	05/17/25 3:54
Open	#43	Incident: privilege_escalation_summary	H	Tasks, Observables, TTPs	Fah	05/17/25 3:54

Image 6: Privilege Escalation Incident in TheHive

The screenshot displays the TheHive web interface, showing the details of a specific incident escalation case. The top navigation bar is the same as in Image 5. The sidebar on the left shows 'Case # 43 - Incident: privilege_escalation_summary'. The main area shows the details of this case, including 'Basic Information' and 'Description'. The 'Basic Information' section includes fields for Title, Severity, TLP, PAP, Assignee, Date, and Tags. The 'Description' section contains a detailed text description of the incident. On the right side, there's a detailed view of the selected case, showing a description of the incident and its associated data.

Case # 43 - Incident: privilege_escalation_summary

Added by Fah 05/17/25 3:54 a minute

Sharing (0) | Close | Flag | Merge | Remove | Responders

Basic Information

Title: Incident: privilege_escalation_summary

Severity: H

TLP: TLP-AMBER

PAP: PAP-AMBER

Assignee: Fah

Date: 05/17/25 3:54

Tags: privilege_escalation_summary

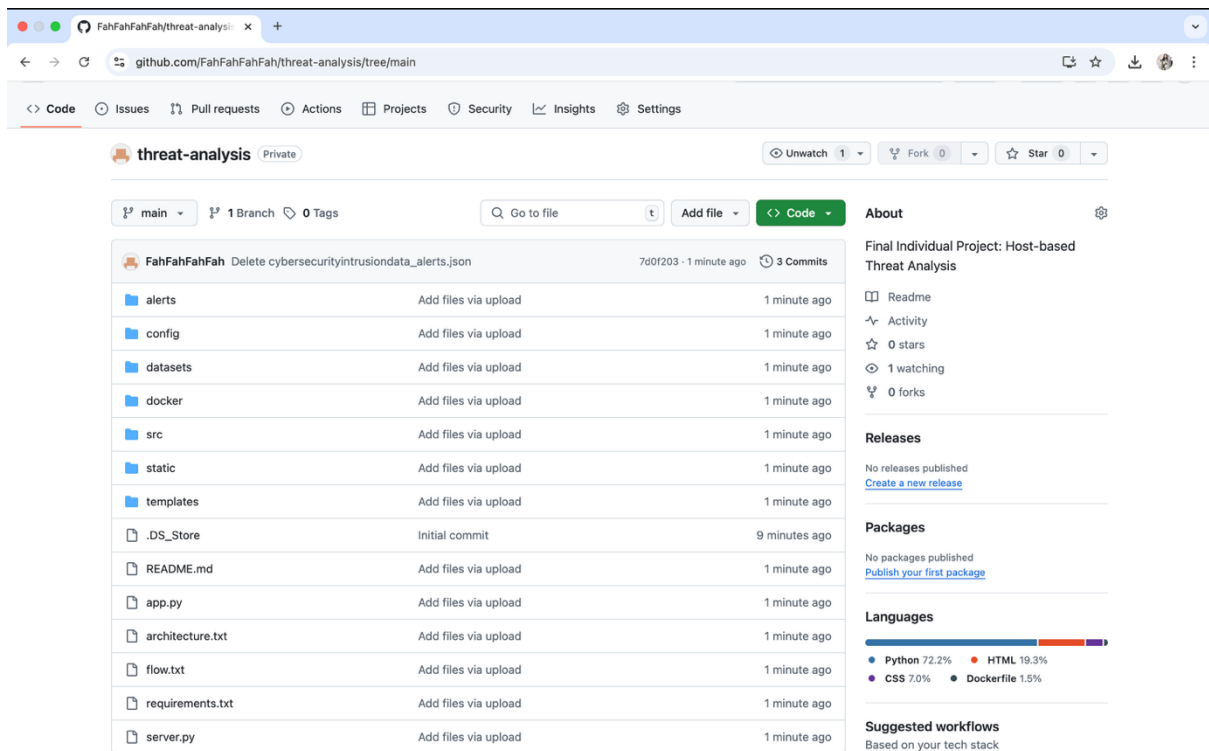
Additional information Layout

No additional information have been specified

Description

{'type': 'privilege_escalation_summary', 'event': {'summary': True, 'count': 1889, 'description': 'Detected 1889 privilege escalation attempts', 'session_ids': ['SID_00004', 'SID_00006', 'SID_00008', 'SID_00009', 'SID_00013', 'SID_00027', 'SID_00030', 'SID_00033', 'SID_00035', 'SID_00036'], 'sample': {'session_id': 'SID_00004', 'network_packet_size': 804, 'protocol_type': 'UDP', 'login_attempts': 4, 'session_duration': 601.2488351708328, 'encryption_used': 'DES', 'ip_reputation_score': 0.1232671757524846, 'failed_logins': 0, 'browser_type': 'Unknown', 'unusual_time_access': 0, 'attack_detected': 1}}, 'severity': 3}

Image 7: Source code uploaded to Github as a repository



Appendix 2: Source Code

1. Web Application Files (Flask Application)

App.py

The main Flask application serving as the interface for file uploads, triggering threat analysis, and connecting integrations (Elasticsearch, Kibana, TheHive).

```
from flask import Flask, request, redirect, url_for, render_template
import os
from server import main as analyze_threats
import requests
import json

app = Flask(__name__)
DATASET_FOLDER = 'datasets' # Use the existing datasets folder
app.config['DATASET_FOLDER'] = DATASET_FOLDER

# Ensure the dataset folder exists
os.makedirs(DATASET_FOLDER, exist_ok=True)
```

```

@app.route('/')
def index():
    return render_template('index.html')

@app.route('/upload', methods=['POST'])
def upload_file():
    if 'file' not in request.files:
        return "No file part", 400
    file = request.files['file']
    if file.filename == '':
        return "No selected file", 400
    if file:
        file_path = os.path.join(app.config['DATASET_FOLDER'], file.filename)
        file.save(file_path)
        # Redirect to analysis page with the uploaded file path
        return redirect(url_for('analyze', file_path=file_path))

@app.route('/analyze')
def analyze():
    file_path = request.args.get('file_path')
    if not file_path:
        return "No file provided for analysis", 400
    try:
        analyze_threats(file_path)
        # Fetch alerts from Elasticsearch and save to alerts.json
        es_url = "http://elasticsearch:9200/threat-alerts/_search?pretty"
        response = requests.get(es_url)
        response.raise_for_status()
        alerts_data = response.json()
        os.makedirs('alerts', exist_ok=True)
        dataset_name = os.path.splitext(os.path.basename(file_path))[0]
        alerts_file = f'alerts/{dataset_name}_alerts.json'
        with open(alerts_file, 'w') as f:
            json.dump(alerts_data, f, indent=2)
    except Exception as e:
        print(f"Error during analysis: {e}")
        return f"Internal Server Error: {e}", 500
    return render_template('analyze.html', file_path=file_path)

@app.route('/elasticsearch-raw')
def elasticsearch_raw():
    return redirect("http://localhost:9200") # Update with your Elasticsearch URL

@app.route('/elasticsearch')
def elasticsearch():
    try:
        es_url = "http://elasticsearch:9200/threat-alerts/_search?pretty"
        response = requests.get(es_url)
        response.raise_for_status()
        alerts_data = response.json()
        return render_template('elasticsearch_results.html', alerts=alerts_data)

```

```

except Exception as e:
    return f"Error loading alerts from Elasticsearch: {e}", 500

@app.route('/kibana')
def kibana():
    return redirect("http://localhost:5601") # Update with your Kibana URL

@app.route('/thehive')
def thehive():
    return redirect("http://localhost:9000") # Update with your TheHive URL

if __name__ == '__main__':
    app.run(host="0.0.0.0", port=5000)

```

Server.py

Core analysis script performing log ingestion, alert detection, exporting alerts to Elasticsearch, and escalating critical alerts to TheHive.

```

from src.log_ingestor import load_logs
from src.rules.failed_login_rules import detect_failed_logins
from src.rules.privilege_escalation_rules import detect_privilege_escalation
from src.elastic_exporter import export_alerts, create_index_template
from src.utils.config_loader import get_config
from src.thehive_exporter import send_case_to_thehive # <-- Add this import

def main(file_path=None):
    # Load configuration
    config = get_config()
    if file_path:
        config['log']['file_path'] = file_path
        log_file = config['log']['file_path']

    # Ensure Elasticsearch index template exists
    create_index_template(config)

    # Load logs from the dataset
    logs = load_logs(log_file)

    # Run detection rules
    failed_login_alerts = detect_failed_logins(logs, config)
    privilege_escalation_alerts = detect_privilege_escalation(logs, config)

    # Combine all alerts
    all_alerts = failed_login_alerts + privilege_escalation_alerts

    # Export to Elasticsearch
    export_alerts(all_alerts, config)

    # Escalate critical alerts to TheHive (example: privilege escalation)

```

```

# Escalate critical alerts to TheHive
if privilege_escalation_alerts:
    # Send a summary case first
    summary = {
        "type": "privilege_escalation_summary",
        "event": {
            "summary": True,
            "count": len(privilege_escalation_alerts),
            "description": f"Detected {len(privilege_escalation_alerts)}
privilege escalation attempts",
            "session_ids": [alert["event"].get("session_id", "unknown") for
alert in privilege_escalation_alerts[:10]],
            "sample": privilege_escalation_alerts[0]["event"] if
privilege_escalation_alerts else {}
        },
        "severity": 3 # Higher severity for the summary
    }
    send_case_to_thehive(summary, config)
    print(f"[+] Sent summary case to TheHive:
{len(privilege_escalation_alerts)} privilege escalation alerts")

    # Then send individual cases (limit to first 5 to avoid flooding)
    for i, alert in enumerate(privilege_escalation_alerts[:5]):
        send_case_to_thehive(alert, config)
        print(f"[+] Sent individual case {i+1}/5 to TheHive:
{alert['event'].get('session_id', 'unknown')}")

```

2. Utility Functions, Config Loaders, Configuration Files

config/config.yaml

Configuration files setting up services and connection details within the Docker environment.

```

elastic:
  host: "elasticsearch"
  port: 9200
  index: "threat-alerts"
  template:
    settings:
      number_of_shards: 1
      number_of_replicas: 0
      lifecycle:
        name: threat-alerts-policy
  mappings:
    mappings:
      properties:
        "@timestamp":
          type: "date"

```

```

        "type":
          type: "keyword"
        "event":
          type: "object"
          dynamic: true
        "severity":
          type: "integer"

log:
  file_path: "/app/datasets/cybersecurity_intrusion_data.csv"

thehive:
  url: "http://thehive:9000"
  api_key: "RBXsyfjzTUbXkVE3Iptx5J7Fhzun+J/U"

thresholds:
  failed_login_attempts: 5

```

config/elastic_config.json

```

{
  "host": "elasticsearch",
  "port": 9200,
  "index": "threat-alerts"
}

```

config/elasticsearch.yml

```

cluster.name: "threat-analysis-cluster"
node.name: "threat-analysis-node"
network.host: 0.0.0.0
http.port: 9200
discovery.type: single-node
xpack.security.enabled: false

```

config/kibana.yml

```

server.name: kibana
server.host: "0.0.0.0"
server.port: 5601
elasticsearch.hosts: ["http://elasticsearch:9200"]
monitoring.ui.container.elasticsearch.enabled: true

```

config/thehive/application.conf

```

play.http.secret.key="okCKogZSl+j9iHf2mG2vL22V4PrB4VV70IJbcdf10DA="
play.http.parser.maxMemoryBuffer: 1G

# JanusGraph configuration

```

```
db {
  provider: janusgraph
  janusgraph {
    storage {
      backend: berkeleyje
      directory: /opt/thp/thehive/db
    }
    index {
      search {
        directory: /opt/thp/thehive/index
      }
    }
  }
}
```

src/utils/config_loader.py

```
import yaml

def get_config(path='config/config.yaml'):
    with open(path, 'r') as f:
        return yaml.safe_load(f)
```

3. Docker Setup

docker/docker-compose.yml

Defines and configures multi-container Docker services, including Flask, Elasticsearch, Kibana, and TheHive.

```
services:

  flask-app:
    build:
      context: ..
      dockerfile: docker/Dockerfile
    ports:
      - "5050:5000" # Map Flask app to localhost:5000
    volumes:
      - ../datasets:/app/datasets # Mount datasets folder
      - ../alerts:/app/alerts
    depends_on:
      - elasticsearch
      - kibana
      - thehive
    networks:
      - threat-analysis-network

  elasticsearch:
```

```

    image: docker.elastic.co/elasticsearch/elasticsearch:8.10.2
    container_name: elasticsearch
    environment:
      - discovery.type=single-node
      - xpack.security.enabled=false
      - ES_JAVA_OPTS=-Xms2g -Xmx2g
    ports:
      - "9200:9200"
    volumes:
      -
    ..../config/elasticsearch.yml:/usr/share/elasticsearch/config/elasticsearch.yml
    networks:
      - threat-analysis-network

kibana:
  image: docker.elastic.co/kibana/kibana:8.10.2
  container_name: kibana
  environment:
    - ELASTICSEARCH_HOSTS=http://elasticsearch:9200
    -
  ELASTIC_SERVICE_TOKEN=AAEAAWVsYXN0aWMva2liYW5hL2RlZmF1bHQ60XLJWEZJVDdSMWFSX0cWdTI2b
  0V5UQ
  ports:
    - "5601:5601"
  volumes:
    - ../config/kibana.yml:/usr/share/kibana/config/kibana.yml
  depends_on:
    - elasticsearch
  networks:
    - threat-analysis-network

thehive:
  image: thehiveproject/thehive4:latest
  ports:
    - "0.0.0.0:9000:9000"
  environment:
    - MAX_HEAP_SIZE=1G
    - HEAP_NEWSIZE=1G
    - APPLICATION_SECRET=okCKogZSl+j9iHf2mG2vL22V4PrB4VV70IJbcdfl0DA=
  volumes:
    - ../config/thehive/application.conf:/etc/thehive/application.conf
    - ./vol/thehive/application.conf:/etc/thehive/application.conf
    - ./vol/thehive/db:/opt/thp/thehive/db
    - ./vol/thehive/index:/opt/thp/thehive/index
    - ./vol/thehive/data:/opt/thp/thehive/data
  depends_on:
    - elasticsearch
  networks:
    - threat-analysis-network

networks:

```



```
threat-analysis-network:
  driver: bridge
```

docker/Dockerfile

Builds the Flask application container, installing dependencies from the provided requirements.txt file.

```
# Dockerfile
FROM python:3.9-slim

# Set working directory
WORKDIR /app

COPY requirements.txt .
RUN pip install --upgrade pip
# Install dependencies
RUN pip install -r requirements.txt
RUN apt-get update && apt-get install -y curl

# Copy everything else (e.g. app.py, server.py, templates/, static/)
COPY . .

# Expose port if needed
EXPOSE 5000

# Run your app
CMD ["python", "app.py"]
```

4. Detection and Analysis Rules

src/rules/failed_login_rules.py

Python script implementing detection logic for identifying repeated failed login attempts.

```
def detect_failed_logins(logs, config):
    alerts = []
    threshold = config['thresholds']['failed_login_attempts']

    for event in logs:
        try:
            failed_logins = int(event.get('failed_logins', 0))
            unusual_time_access = int(event.get('unusual_time_access', 0))
            attack_detected = int(event.get('attack_detected', 0))

            # Check if this meets failed login criteria but NOT privilege
            escalation criteria
            is_failed_login = failed_logins >= threshold
```

```

        is_privilege_escalation = unusual_time_access == 1 or (attack_detected
== 1 and failed_logins == 0)

        if is_failed_login and not is_privilege_escalation:
            alerts.append({"type": "failed_login", "event": event})
        except (ValueError, TypeError):
            continue

    return alerts

```

src/rules/privilege_escalation_rules.py

Detection logic for identifying events indicative of privilege escalation.

```

def detect_privilege_escalation(logs, config):
    alerts = []
    for event in logs:
        try:
            failed_logins = int(event.get('failed_logins', 0))
            unusual_time_access = int(event.get('unusual_time_access', 0))
            attack_detected = int(event.get('attack_detected', 0))

            # Check if this satisfies privilege escalation criteria
            is_privilege_escalation = unusual_time_access == 1 or (attack_detected
== 1 and failed_logins == 0)

            if is_privilege_escalation:
                # This is a privilege escalation event
                alerts.append({"type": "privilege_escalation", "event": event})
        except (ValueError, TypeError):
            continue

    return alerts

```

src/alert_engine.py

```

import re

def detect_failed_logins(logs, config):
    alerts = []
    threshold = config['thresholds']['failed_login_attempts']
    failed_login_pattern = re.compile(
        r'(failed password|authentication error|pam: authentication failure|illegal
user|invalid user)',
        re.IGNORECASE
    )

    for event in logs:

```

```

        message = event.get('message', '')
        if failed_login_pattern.search(message):
            alerts.append({"type": "failed_login", "event": event})
        try:
            if int(event.get('failed_logins', 0)) >= threshold:
                alerts.append({"type": "failed_login", "event": event})
        except (ValueError, TypeError):
            continue
    return alerts

def detect_privilege_escalation(logs, config):
    alerts = []
    privilege_pattern = re.compile(
        r'(sudo|su|setuid|chmod 777|root access|session opened for user root|not in sudoers)',
        re.IGNORECASE
    )

    for event in logs:
        message = event.get('message', '')
        if privilege_pattern.search(message):
            alerts.append({"type": "privilege_escalation", "event": event})
    return alerts

```

5. Log Decoders

src/decoders/pam_decoder.py

```

import re

def decode_pam_line(line: str) -> dict:
    """
    Extract fields from PAM log lines.
    """
    fields = {}
    pam_regexes = {
        'user': r'user=(\S+)',
        'srcuser': r'ruser=(\S+)',
        'srcip': r'rhost=(\S+)',
        'logname': r'logname=(\S+)',
        'uid': r'uid=(\S+)',
        'euid': r'euid=(\S+)',
        'tty': r'tty=(\S+)',
    }
    for field, pattern in pam_regexes.items():
        match = re.search(pattern, line)
        if match:
            fields[field] = match.group(1)
    fields['message'] = line
    return fields

```

src/decoders/ssh_decoder.py

```
import re

def decode_ssh_line(line: str) -> dict:
    """
    Extract fields from SSH log lines.
    """
    patterns = [
        (r'Failed password for (invalid user )?(\S+) from (\S+)', ['user',
'srcip']),
        (r'Invalid user (\S+) from (\S+)', ['user', 'srcip']),
        (r'Accepted password for (\S+) from (\S+)', ['user', 'srcip']),
        (r'User (\S+) from (\S+)', ['user', 'srcip']),
    ]
    for pattern, fields in patterns:
        match = re.search(pattern, line)
        if match:
            values = match.groups()
            return {k: v for k, v in zip(fields, values) if v}
    return {'message': line}
```

src/decoders/su_decoder.py

```
import re

def decode_su_line(line: str) -> dict:
    """
    Extract SU user transitions or failures.
    """
    patterns = [
        (r"FAILED su for (\S+) by (\S+)", ['dstuser', 'srcuser']),
        (r"BAD SU (\S+) to (\S+)", ['srcuser', 'dstuser']),
        (r"^su (\S+)' .* for (\S+) on", ['dstuser', 'srcuser']),
    ]
    for pattern, fields in patterns:
        match = re.search(pattern, line)
        if match:
            values = match.groups()
            return {k: v for k, v in zip(fields, values)}
    return {'message': line}
```

src/decoders/sudo_decoder.py

```
import re

def decode_sudo_line(line: str) -> dict:
    """
    Extract sudo usage data including srcuser, dstuser, command.
    """
    fields = {}
```

```

# Match username before colon
user_match = re.search(r'^\s*(\S+)\s*:', line)
if user_match:
    fields['srcuser'] = user_match.group(1)
# Extract standard sudo fields
for field, pattern in {
    'tty': r'TTY=(\S+)',
    'pwd': r'PWD=(\S+)',
    'dstuser': r'USER=(\S+)',
    'command': r'COMMAND=(.+)',
}.items():
    match = re.search(pattern, line)
    if match:
        fields[field] = match.group(1)
fields['message'] = line
return fields

```

6. Elasticsearch and Alert Exporting

src/elastic_exporter.py

Python script handling alert exportation to Elasticsearch, including the setup of index templates and data sanitization.

```

import requests
import math

def sanitize_json(obj):
    if isinstance(obj, dict):
        return {k: sanitize_json(v) for k, v in obj.items()}
    elif isinstance(obj, list):
        return [sanitize_json(v) for v in obj]
    elif isinstance(obj, float):
        if math.isnan(obj) or math.isinf(obj):
            return None
        return obj
    else:
        return obj

def create_index_template(config):
    url =
f"http://{config['elastic']['host']}:{config['elastic']['port']}/_index_template/threat-alerts-template"
    template = {
        "index_patterns": [f"{config['elastic']['index']}*"],
        "template": {
            "settings": {
                "number_of_shards": 1,
                "number_of_replicas": 0,

```

```

        "lifecycle": {
            "name": "threat-alerts-policy"
        }
    },
    "mappings": {
        "properties": {
            "@timestamp": {"type": "date"},
            "type": {"type": "keyword"},
            "event": {"type": "object"},
            "severity": {"type": "integer"}
        }
    }
}

response = requests.put(url, json=template)
response.raise_for_status()
print("[+] Index template created successfully.")

def export_alerts(alerts, config):
    url =
f"http://{config['elastic']['host']}:{config['elastic']['port']}/{config['elastic']
['index']}/_doc"
    for alert in alerts:
        sanitized_alert = sanitize_json(alert)
        response = requests.post(url, json=sanitized_alert)
        response.raise_for_status()
    print(f"[+] Exported {len(alerts)} alerts to Elasticsearch.")

```

7. Data Processing and ETL

src/log_ingestor.py

Loads log files (CSV format) into structured data for processing.

```

import pandas as pd

def load_logs(file_path):
    print(f"[+] Loading logs from {file_path}")
    return pd.read_csv(file_path).to_dict(orient='records')

```

enhanced_etl.py

Performs enhanced ETL processes, including log enrichment and risk scoring.

```

import pandas as pd
import json
import re
import ipaddress
import requests

```

```

from datetime import datetime

class EnhancedLogIngestor:
    """Enhanced ETL pipeline supporting multiple log formats and basic
    enrichment"""

    def __init__(self, config):
        self.config = config
        # Cache for IP enrichment data to avoid repeated lookups
        self.ip_cache = {}

    def load_logs(self, source):
        """Universal log loader supporting multiple formats"""
        print(f"[+] Loading logs from {source['path']} (format:
{source['format']})")

        if source['format'] == 'csv':
            return self._load_csv(source['path'])
        elif source['format'] == 'json':
            return self._load_json(source['path'])
        elif source['format'] == 'syslog':
            return self._load_syslog(source['path'])
        else:
            print(f"[!] Unsupported format: {source['format']}")
            return []

    def _load_csv(self, path):
        """Load and parse CSV file"""
        return pd.read_csv(path).to_dict(orient='records')

    def _load_json(self, path):
        """Load and parse JSON file"""
        with open(path, 'r') as f:
            return json.load(f)

    def _load_syslog(self, path):
        """Load and parse syslog-format file"""
        events = []
        syslog_pattern = re.compile(
            r'(\w{3})\s+\d+\s+\d+:\d+:\d+\s+(\S+)\s+(\S+)(?:\[(\d+)\])?:\s+(.*)'
        )

        with open(path, 'r') as f:
            for line in f:
                match = syslog_pattern.match(line.strip())
                if match:
                    timestamp, host, service, pid, message = match.groups()
                    events.append({
                        'timestamp': timestamp,
                        'host': host,
                        'service': service,

```

```

        'pid': pid,
        'message': message
    })
    return events

def enrich_events(self, events):
    """Add enrichment data to events"""
    enriched = []

    for event in events:
        # Make a copy to avoid modifying the original
        enriched_event = event.copy()

        # Normalize timestamp if present but in different formats
        if 'timestamp' in enriched_event:
            enriched_event['@timestamp'] =
self._normalize_timestamp(enriched_event['timestamp'])

        # Add source IP enrichment if available
        source_ip = enriched_event.get('source_ip')
        if source_ip and self._is_valid_ip(source_ip):
            geo_data = self._get_ip_geo(source_ip)
            if geo_data:
                enriched_event['geo'] = geo_data

        # Add simple risk score based on available indicators
        enriched_event['risk_score'] =
self._calculate_risk_score(enriched_event)

        enriched.append(enriched_event)

    return enriched

def _normalize_timestamp(self, timestamp):
    """Convert various timestamp formats to ISO standard"""
    # This is a simplified example – production would handle many formats
    try:
        if isinstance(timestamp, str):
            # Try a few common formats
            formats = [
                '%Y-%m-%d %H:%M:%S',
                '%b %d %H:%M:%S',
                '%d/%b/%Y:%H:%M:%S'
            ]

            for fmt in formats:
                try:
                    dt = datetime.strptime(timestamp, fmt)
                    return dt.isoformat()
                except ValueError:
                    continue
    
```



```

        return timestamp # Return original if conversion failed
    except Exception:
        return timestamp

def _is_valid_ip(self, ip):
    """Check if string is valid IP address"""
    try:
        ipaddress.ip_address(ip)
        return True
    except ValueError:
        return False

def _get_ip_geo(self, ip):
    """Get geolocation data for IP (with caching)"""
    if ip in self.ip_cache:
        return self.ip_cache[ip]

    # This would normally use a real geo-IP service
    # For demo, just return mock data
    mock_geo = {
        "country_code": "US",
        "country_name": "United States",
        "city": "Phoenix",
        "latitude": 33.4484,
        "longitude": -112.0740
    }

    self.ip_cache[ip] = mock_geo
    return mock_geo

def _calculate_risk_score(self, event):
    """Calculate simple risk score based on event attributes"""
    score = 0

    # Use all available indicators
    if event.get('failed_logins', 0) >= 5:
        score += 20

    if event.get('unusual_time_access', 0) == 1:
        score += 30

    if event.get('attack_detected', 0) == 1:
        score += 25

    # Additional indicators
    if 'message' in event:
        message = event['message'].lower()
        if 'root' in message or 'admin' in message:
            score += 10

```

```

        if 'firewall' in message and 'block' in message:
            score += 15

    # IP reputation if available
    ip_reputation = event.get('ip_reputation_score', 0)
    if isinstance(ip_reputation, (int, float)) and ip_reputation > 0.5:
        score += 15

    return score

# Example usage
if __name__ == "__main__":
    # Demo
    config = {"thresholds": {"failed_login_attempts": 5}}
    ingestor = EnhancedLogIngestor(config)

    # Load from different sources
    csv_events = ingestor.load_logs({"format": "csv", "path":
"datasets/cybersecurityintrusiondata.csv"})

    # Enrich events
    enriched = ingestor.enrich_events(csv_events[:5]) # Process first 5 for demo

    # Show results
    for event in enriched:
        print(f"Event: {event.get('session_id', 'unknown')} | Risk Score:
{event['risk_score']}")
        if 'geo' in event:
            print(f"    Location: {event['geo']['city']},
{event['geo']['country_name']}")

```

8. TheHive Integration

src/thehive_exporter.py

Exports detected critical incidents to TheHive, creating automated cases for the SOC team's investigation.

```

import requests

def send_case_to_thehive(alert, config):
    url = f"{config['thehive']['url'].rstrip('/')}/api/case"
    headers = {
        "Authorization": f"Bearer {config['thehive']['api_key']}",
        "Content-Type": "application/json"
    }
    case = {
        "title": f"Incident: {alert['type']}",
        "description": str(alert),
        "severity": alert.get("severity", 2),
    }

```

```

        "tlp": 2,
        "tags": [alert['type']],
    }
    response = requests.post(url, json=case, headers=headers)
    response.raise_for_status()
    print(f"[+] Sent alert to TheHive as a case: {case['title']}")

```

9. Frontend (HTML, CSS, JS)

static/css/styles.css

CSS for UI/UX styling

```

/* Reset & Layout */
html, body {
    height: 100%;
    margin: 0;
    padding: 0;
    font-family: Arial, sans-serif;
    background-color: #f4f4f9;
    color: #333;
    display: flex;
    flex-direction: column;
}

/* Header */
header {
    text-align: center;
    padding: 20px;
    background-color: #0073e6;
    color: white;
}

/* Main */
main {
    flex: 1;
    padding: 20px;
    max-width: 900px;
    margin: 0 auto;
}

/* Footer */
footer {
    text-align: center;
    padding: 10px;
    background-color: #333;
    color: white;
}

/* Upload Section */

```

```

.upload-section {
  text-align: center;
  margin-bottom: 50px;
}

.upload-section h2 {
  font-size: 2em;
  margin-bottom: 20px;
}

.upload-section form {
  display: flex;
  flex-wrap: wrap;
  justify-content: center;
  align-items: center;
  gap: 20px;
}

.upload-section label {
  font-size: 1.2em;
}

.upload-section input[type="file"] {
  font-size: 1em;
}

.upload-section button {
  background-color: #0073e6;
  color: white;
  border: none;
  padding: 12px 24px;
  font-size: 1.1em;
  cursor: pointer;
  border-radius: 5px;
  height: 42px;
  display: flex;
  align-items: center;
}

.upload-section button:hover {
  background-color: #005bb5;
}

/* Integration Section */
.integrations-section {
  text-align: center;
}

.integrations-section h2 {
  font-size: 2em;
  margin-bottom: 20px;
}

```

```

}

.integration-links {
  display: flex;
  justify-content: center;
  gap: 40px;
  flex-wrap: wrap;
}

.integration-links a {
  text-decoration: none;
  color: #333;
  text-align: center;
  width: 150px;
}

.integration-links img {
  width: 100px;
  height: 100px;
  object-fit: contain;
  margin: 0 auto 10px;
  display: block;
}

.integration-links span {
  font-size: 1em;
  font-weight: bold;
  display: block;
}

```

templates/index.html

Web interface templates

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Threat Analysis Dashboard</title>
  <link rel="stylesheet" href="{% url_for('static', filename='css/styles.css') %}">
</head>
<body>
  <header>
    <h1>Threat Analysis Dashboard</h1>
    <p>Analyze threats, visualize results in Kibana and Elasticsearch, and
    automate case creation in TheHive.</p>
  </header>

```

```

<main>
  <section class="upload-section">
    <h2>Upload Dataset</h2>
    <form action="/upload" method="post" enctype="multipart/form-data">
      <label for="file">Upload Dataset (CSV):</label>
      <input type="file" name="file" id="file" required>
      <button type="submit">Upload and Analyze</button>
    </form>
  </section>

  <section class="integrations-section">
    <h2>Integrations</h2>
    <div class="integration-links">
      <a href="/elasticsearch-raw" target="_blank">
        
        <span>Elasticsearch</span>
      </a>
      <a href="/kibana" target="_blank">
        
        <span>Kibana</span>
      </a>
      <a href="/thehive" target="_blank">
        
        <span>TheHive</span>
      </a>
    </div>
  </section>
</main>

<footer>
  <p>&copy; 2025 Threat Analysis Platform. All rights reserved.</p>
</footer>
</body>
</html>

```

templates/analyze.html

Web interface templates

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Analysis Complete</title>
  <link rel="stylesheet" href="{{ url_for('static', filename='css/styles.css')
}}">
</head>
<body>

```

```

<header>
  <h1>Threat Analysis Dashboard</h1>
  <p>Analyze threats, visualize results in Kibana and Elasticsearch, and
automate case creation in TheHive.</p>
</header>

<main>
  <section class="analysis-complete">
    <h2>Analysis Complete</h2>
    <p>The dataset has been successfully analyzed.</p>
    <p>File analyzed: <strong>{{ file_path.split('/')[ -1] }}</strong></p>
  </section>

  <section class="results-section">
    <h2>View Results</h2>
    <div class="integration-links">
      <a href="/elasticsearch" target="_blank">
        
        <span>See Elasticsearch Alerts</span>
      </a>
      <a href="/kibana" target="_blank">
        
        <span>Kibana</span>
      </a>
      <a href="/thehive" target="_blank">
        
        <span>TheHive</span>
      </a>
    </div>
  </section>
</main>

<footer>
  <p>&copy; 2025 Threat Analysis Platform. All rights reserved.</p>
</footer>
</body>
</html>

```

templates/elasticsearch_results.html

Web interface templates

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Elasticsearch Alerts</title>
  <style>

```

```

    body { font-family: Arial, sans-serif; background: #f7f7f7; margin: 0;
padding: 2em; }
    h1 { color: #333; }
    .cards-container { display: flex; flex-wrap: wrap; gap: 1em; }
    .alert-card {
        background: #fff;
        border-radius: 8px;
        box-shadow: 0 2px 8px #ccc;
        padding: 1em;
        min-width: 300px;
        max-width: 350px;
        margin-bottom: 1em;
        flex: 1 1 300px;
    }
    .alert-type { font-weight: bold; color: #0078d4; margin-bottom: 0.5em; }
    .alert-field { margin-bottom: 0.2em; }
    .back-btn { display: inline-block; margin-top: 2em; padding: 0.5em 1em;
background: #0078d4; color: #fff; border: none; border-radius: 4px; text-
decoration: none; }
    .back-btn:hover { background: #005fa3; }
</style>
</head>
<body>
    <h1>Elasticsearch Alerts</h1>
    <div class="cards-container">
        {% for hit in alerts.hits.hits %}
        <div class="alert-card">
            <div class="alert-type">{{ hit._source.type|capitalize }}</div>
            {% for key, value in hit._source.event.items() %}
                <div class="alert-field"><strong>{{ key }}:</strong> {{ value
}}</div>
            {% endfor %}
        </div>
        {% endfor %}
    </div>
    <a href="javascript:history.back()" class="back-btn">Back to Analysis</a>
</body>
</html>

```